

UNIVERSITY OF CALIFORNIA

SANTA CRUZ

**True to Form:
Rising and Falling Declaratives as Questions in English**

A dissertation submitted in partial satisfaction
of the requirements for the degree of

DOCTOR OF PHILOSOPHY

in

LINGUISTICS

by

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December 2001

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Abstract

True to Form: Rising and Falling Declaratives as Questions in English

Christine Gunlogson

This dissertation is concerned with the meaning and use of two kinds of declarative sentences:

- (1) It's raining?
- (2) It's raining.

The difference between (1) and (2) is intonational: (1) has a final rise (indicated by the question mark), while (2) ends with a fall.

The central claim of the thesis is that the meaning and use of both kinds of sentences must be understood in terms of the meaning of their defining formal elements, namely declarative sentence type and rising vs. falling intonation. I support that claim through an investigation of the use of declaratives as questions. On the one hand, I demonstrate that rising and falling declaratives share an aspect of conventional meaning attributable to their declarative form, distinguishing them both from the corresponding polar interrogative (*Is it raining?*) and constraining their use as questions. On the other hand, since (1) and (2) constitute a minimal pair, differing only in intonation, systematic differences in character and function between them – in particular, the relative ‘naturalness’ of (1) as a question compared to (2) – must be located in the contrast between the fall and the rise.

To account for these two sets of differences, I give a compositional account of rising and falling declaratives under which declarative form (in contrast to interrogative) expresses commitment to the propositional content of the declarative. Rising vs. falling intonation on declaratives is responsible for attribution of the commitment to the Addressee vs. the Speaker, respectively. The result is an inherent contextual ‘bias’ associated with declaratives, which constitutes the crucial point of difference with interrogatives. The compositional analysis is implemented in the framework of context update semantics (Heim 1982 and others), using an articulated version of the Common Ground (Stalnaker 1978) that distinguishes the commitments of the individual discourse participants.

Restrictions on the use of declaratives as questions, as well as differences between rising and falling declaratives as questions, are shown to follow from this account. I argue that neither rising nor falling declaratives are inherently questioning – rather, the questioning function of declaratives arises through the interaction of sentence type, intonation, and context.

Acknowledgments

This thesis would not have been possible without the wise guidance and support of my advisor, Donka Farkas, as well as the rest of my committee: Daniel Büring, Bill Ladusaw, and Geoff Pullum. As a group and individually, they encouraged me to pursue my ideas while insisting that those ideas be made formally precise. They were unfailingly generous with time, with thoughtful criticism, and with enthusiasm about the project. I especially thank Bill, Geoff, and Donka for their early input, which taught me the importance of framing the problem empirically and started me on the search for relevant data. At later stages, Donka and Daniel were extremely generous with their time, spending countless hours puzzling over problems with me, reading notes and drafts, and critiquing various proposals. I feel fortunate to have worked with such a committee, and I am tremendously grateful to my chair, Donka Farkas, who has been the best of advisors.

I feel fortunate as well to have earned my Ph.D. in a department with a strong sense of community and dedication to both teaching and scholarship. I'd like to thank the members of that community – faculty, students, and staff – for their part in making it such a good place to do linguistics. Special thanks to Bill Ladusaw and Armin Mester for teaching the seminars that first got me interested in this topic, and to Jaye Padgett for his help in the lab.

Finally, I thank my friends and colleagues for their patience, and I promise to do my best to break the habit of pointing out rising declaratives at every turn.

Chapter 1

Introduction

1.1 Overview

Consider the three sentences in (3):

- | | | |
|-----|-------------------|-----------------------------------|
| (3) | a. Is it raining? | <i>Rising polar interrogative</i> |
| | b. It's raining? | <i>Rising declarative</i> |
| | c. It's raining. | <i>Falling declarative</i> |

(3a) is a polar interrogative, the prototypical way to ask a polar (yes/no) question.

(3c) is a declarative with falling intonation, the canonical way to make a statement.

The declarative with rising intonation, indicated by the question mark in (3b), is superficially more similar in function to (3a) than (3c). Intuitively, the rise seems to impart the force of a question to what would otherwise be naturally interpreted as a statement. Thus, a familiar use of rising declaratives is as a kind of polar question, much like the corresponding syntactic interrogative:

- | | | | |
|-----|---------------------------------------|---|-------------------------------------|
| (4) | a. That's a persimmon? | ≈ | Is that a persimmon? |
| | b. You're leaving for vacation today? | ≈ | Are you leaving for vacation today? |

But the story of rising declarative questions cannot be as simple as the rough equivalence in (4) suggests. It turns out that rising declaratives as questions are subject to contextual restrictions that do not apply to their interrogative counterparts.

For example, declarative questions are not appropriate in situations where the questioner is supposed to be impartial or uninformed, as in a courtroom or committee hearing:

- (5) [at a committee hearing]
- a. Are you a member of the Communist party?
 - b. #You're a member of the Communist party?
 - c. #You're a member of the Communist party.

Furthermore, rising declarative questions cannot be used 'out of the blue', without any relevant preceding context, as interrogatives can be. The interrogative in (6a) is felicitous as an initial remark, without any preceding discussion of persimmons, while the rising declarative in (6b) is odd in the same circumstances:

- (6) [to coworker eating a piece of fruit]
- a. Is that a persimmon?
 - b. #That's a persimmon?
 - c. #That's a persimmon.

The falling declaratives in (5c)-(6c) are also unacceptable as questions in these circumstances, a fact that is intuitively unsurprising but nonetheless significant. Given that rising declaratives pattern with their falling declarative counterparts in the above examples, and not with interrogatives, it is reasonable to look to declarative form for an explanation of the constraints on distribution. That is exactly the approach I will take in this dissertation.

At the same time, the intuition that rising declaratives are more suited to questioning than falling ones is undeniable. Evidence supporting that intuition can be found in examples like (7), where the rising declarative patterns with the interrogative:

- (7) A: The king of France is bald.
B's response:
a. Is France a monarchy?
b. France is a monarchy?
c. #France is a monarchy.

Since (7b) and (7c) differ only in their intonational contour, we must look to the difference between the rise and the fall for an explanation of the question-like behavior of rising declaratives. In the account to be given here, the explanation will crucially depend not just upon the meaning of the rise but on the interaction of the rise with the meaning proposed for declarative form.

The primary goal of the thesis is to give a *compositional* analysis of the meaning of the formal elements of rising and falling declaratives – namely sentence type and intonational contour – from which the observed distributional patterns follow. Beyond that, I aim to understand how the meaning of rising and falling declaratives relates to their interpretation as questions in particular contexts. The minimal pair methodology illustrated by (5)-(7) is crucial to the enterprise. Rising declaratives are compared with rising interrogatives on the one hand and with falling declaratives on the other, holding constant the lexical content and location of the nuclear accent. (Interrogatives, too, may be either rising or falling, but only rising interrogatives will be considered in this thesis.) This strategy isolates the separate contributions of sentence type and intonation.

A central task of the thesis is to document systematic restrictions on the distribution of declarative questions across contexts and note empirical reflexes of

the intuitive differences between the use of rising vs. falling declaratives as questions, grounding the analysis in a body of data from which (5)-(7) are drawn. Some of the individual observations have been made before; some are (to my knowledge) novel. The main innovation, however, is organizing them in a way that allows significant generalizations to emerge. These empirically based generalizations – there are three altogether – serve both as the springboard for my analysis and as a relatively concrete standard of evaluation against which my own proposal and others can be judged.

The main body of the analysis consists of 0-Chapter 4. In 0, minimal-pair methodology is first used to examine the distribution of declarative questions vis-à-vis their interrogative counterparts as well as the distribution of rising vs. falling declarative questions. Two of the three major descriptive generalizations emerge in this chapter:

- (8) Declaratives express a bias that is absent with the use of interrogatives; they cannot be used as neutral questions.
- (9) Rising declaratives, like interrogatives, fail to commit the Speaker to their content.

At the end of the chapter a proposal is advanced in informal terms to account for those generalizations. The gist of the proposal is that declaratives convey *commitment* and cannot, therefore, be used as *neutral questions*; they are biased, even when functioning as questions. The difference between a falling and a rising declarative is in the attribution of commitment: a falling declarative commits the Speaker, while the rising version assumes commitment by the Addressee. Because

rising declaratives fail to commit the Speaker, they have a broader range of distribution as questions than falling declaratives – but not as broad as syntactic interrogatives.

Chapter 3 implements the proposal, giving formal substance to the intuitive notions of *commitment*, *neutrality*, and *bias* and linking these categories to sentence type. The analysis has two main components: (i) an articulated representation of the discourse context, founded on Stalnaker's (1978) model of the Common Ground, that distinguishes the contributions of each discourse participant and allows for the characterization of contextual bias and neutrality; (ii) a context-update account of rising and falling declaratives that is fully compositional with respect to the elements of intonation and sentence type. I model the effects of discourse moves as context updates, proposing that rising declaratives have the effect of committing the Addressee to their propositional content. Such a move results in contextual bias, consistent with (8), but does not commit the Speaker one way or another, as required for (9). Falling declaratives do commit the Speaker, and this limits their function as questions relative to rising declaratives. The notion of *commitment* is realized as a property of declarative updates across contexts, which can be summarized in terms of contextual bias and neutrality: declarative updates, unlike interrogative ones, never result in a neutral context.

The analysis of declarative bias offered in Chapter 3 accounts for examples like (5) and (7) above. However, the restriction illustrated in (6), which demands preceding context of a particular sort for declarative questions, does not follow in

any immediately obvious way from the account of declarative bias. In Chapter 4 I argue for characterizing the restriction descriptively as in (10):

- (10) The Contextual Bias Condition: Rising declaratives can only be used as questions in contexts where the Addressee is already publicly committed to the proposition expressed.

The Contextual Bias Condition is derived from a more general principle governing the interpretation of utterances as polar questions: I claim that uninformativeness with respect to the Addressee is a necessary condition for the interpretation of a move as a question. Interrogatives are uninformative by nature and thus can meet the condition in any context. Declaratives, however, can count as questions only if the Addressee is understood to be *already* publicly committed to the proposition expressed by the declarative – that is, only if the Contextual Bias Condition is met. The analysis predicts, correctly, that in addition to their familiar ‘echoing’ function, rising declaratives may be used to question presuppositions and inferences taken to be logical consequences of the Addressee’s public position, whether or not such inference finds its basis in a preceding utterance. Falling declaratives as questions are subject to additional restrictions related to the fact that they commit the Speaker to their content, differing on this point from both rising declaratives and interrogatives.

The notion of ‘question’ to be explicated in this paper is a broad one, not limited to the prototypical case of an ‘information question’. In keeping with the empirical orientation of the paper, I approach the task of characterizing the category of polar questions in a distributional way. It is observed throughout the

thesis that, generally speaking, wherever a rising declarative can be employed as a question, the corresponding rising interrogative is possible as well. Moreover, the range of interpretations possible for such uses is the same for the two sentence types. This convergence suggests a solution to the problem of what constitutes a polar question. The relevant sense of ‘polar question’, I propose, is the general sense in which any felicitous use of a syntactic polar interrogative is intuitively a question, no matter what its intended function in the discourse. An utterance of a sentence of a different type may achieve the status of a polar question in a particular context to the extent that it produces the effect a syntactic interrogative would have in the same context. This approach correctly allows for a wide range of attitudes and discourse goals to be associated with ‘questioning’ in a broad sense. At the same time, it accounts for otherwise puzzling (and heretofore unremarked) distributional facts concerning the distribution and interpretation of declarative questions relative to interrogative ones, as well as the restricted distribution of falling declarative questions relative to rising ones. The goal of expressing our intuitions about the relative ‘naturalness’ of rising and falling declaratives as questions is achieved by defining a notion of relative markedness in distributional terms.

Concentrating on the use of declaratives as questions, as I do throughout the thesis, has the virtue of illuminating the *limits* on what the form can be used to do. As is frequently the case in linguistic investigation, understanding the logic of what patterns are *not* to be found can shed light on the patterns that do exist. Thus, the

conclusions drawn from the study of questioning uses of declaratives have significant implications in other areas, including assertive uses, and consequences as well for interrogative sentence types and their interaction with intonational contours.

The results of the thesis, and some of its implications, are summarized in Chapter 5. The remainder of this chapter covers the fundamental working assumptions and limitations of the thesis (Section 1.2) and offers some preliminary remarks on earlier work (Section 1.3).

1.2 Assumptions

In this section I will give an overview of key assumptions, terminology, and exclusions. As has already been mentioned, the analysis relies throughout on the use of minimal pairs; (3) is repeated below for reference:

- | | | |
|-----|-------------------|-----------------------------------|
| (3) | a. Is it raining? | <i>Rising polar interrogative</i> |
| | b. It's raining? | <i>Rising declarative</i> |
| | c. It's raining. | <i>Falling declarative</i> |

Throughout the thesis I will indicate a final rise with a question mark and a final fall with a period, consistent with ordinary orthographic conventions for declaratives. I am concerned only with a subset of possible intonational contours, as will be detailed below.

In each set of contrasts like (3), there are two minimal pairs. The (a) and (b) cases are to be treated as contrasting only in syntactic form; the (b) and (c) cases

are identical except for intonational contour. The propositional content, or *descriptive content*, as I will often refer to it, is intuitively the same across all three sentences, disregarding the subject-auxiliary inversion characteristic of the polar interrogative in the (a) case. In the context update account offered in Chapter 3, I will indeed treat all three sentence types as having the same propositional ‘core’, with different handling of the core descriptive content depending on the sentence type and intonation. This is a departure from traditional semantic accounts of interrogatives (e.g., Hamblin 1973, Karttunen 1977), but it is a natural development of the context update approach. See Groenendijk 1999 for a recent precedent.

The terms (*polar*) *interrogative* and *declarative* are understood as referring to syntactic sentence types, while (*polar*) *question* is used for the name of a pragmatic category to which utterances of both interrogatives and declaratives can belong. Polar interrogatives as well as declaratives can have either rising or falling intonation, but I restrict attention throughout the thesis to the rising variety. My concern is primarily with declaratives, and for that reason I consider only polar interrogatives that contrast minimally with declaratives. Thus I do not treat alternative questions or *wh*- questions. I also avoid examples with explicitly negated interrogatives, as the interpretation of negated interrogatives is complicated in a way that the interpretation of negated declaratives is not. (Examples illustrating these differences between interrogatives and declaratives are given in Section 2.2.) Finally, in order to achieve the desired minimal pairing with rising declaratives, I will consider only interrogatives with rising intonation – a typical pattern in

American English, though by no means the only possibility. What qualifies as ‘rising’ intonation will be discussed shortly.

A fundamental (and I trust, noncontroversial) premise of this dissertation is that syntactic sentence type does not determine intonational contour. It should be clear from the basic subject matter of the thesis – rising and falling declaratives – that a phrase like ‘the intonation of declaratives’ makes very little sense. As Bolinger 1982 points out, neither does ‘the intonation of non-declaratives’, or for that matter, any phrase of the form ‘the intonation of X’, where X is a grammatical category. The approach I take here, consistent with Ladd 1980, is to treat the rise and the fall as elements of an intonational lexicon. The choice of elements from the intonational lexicon for a particular sentence is, in principle, free (within the limits of what constitutes a well-formed tune) – just as the choice of lexical elements in a sentence is free (within the limits of what constitutes a well-formed sentence). That does not mean every conceivable combination of tune and sentence is expected to be equally well formed. Rather, it means that when we are able to isolate inappropriate combinations of elements, we have a legitimate clue to the semantic properties of the elements involved. The reasoning is no different, in principle, than that employed in semantic investigation of more traditional categories.

Syntactic sentence type doesn’t determine speech act category or illocutionary force, either. This premise, too, is noncontroversial – clearly not all uses of declaratives, even falling declaratives, are assertions, and not all interrogatives function as requests for information. At the same time, I do not want

to deny that there is intuitively *some* natural connection between sentence type and certain discourse functions. (The same point holds for intonational choices.) The challenge is to give an account of declarative meaning that is abstract enough to cover a range of uses but that also provides insight into why some uses seem more central than others. The point I want to emphasize at the outset is that in the account I will offer, declarative sentence type contributes to the *conventional* meaning of the sentence – its locutionary content, not its illocutionary force. Commitment as I will define it here is not a speech act.

The contrast between declarative and interrogative sentence types is straightforward compared to the intonational contrast. The inventory of intonational morphemes in English is not a settled matter. This lack of consensus complicates the problem of approaching intonational meaning. There is little doubt, however, that *some* significant contrast exists between a declarative with prototypical falling (sometimes referred to as ‘declarative’ or ‘statement’) intonation and the same declarative content with rising (‘question’) intonation. Acknowledging the existence of such a basic contrast, the issues have to do with characterizing which contours fall into each category, how the intonational contrasts are represented, and what constitutes the minimal units of analysis from a semantic viewpoint. I will outline a particular set of assumptions below, emphasizing at the outset that my assumptions are intended to facilitate a broad *semantic* investigation and are not intended to be an adequate characterization of the full set of phonological contrasts. My goal in the present work is to identify a robust set of semantic contrasts that I

hypothesize to be attached to the contrast between particular intonational categories, characterized as ‘rising’ and ‘falling’. The assumptions I make about the relevant categories remain rather broad; but if the semantic contrasts identified hold up, they can in turn inform refinement or revision of the phonological representations.

I begin with some basics of terminology. The overall contour associated with a lexical string is determined in part by the nature of the particular pitch accents chosen within the body of the string, but also includes characteristic pitch movements between the final, or ‘nuclear’, pitch accent and the end of the string. It is these ‘post-nuclear’ movements that correspond to what we intuitively think of as a ‘final rise’ or ‘final fall’ at the end of an uttered sentence. The ‘rise’ or ‘fall’ is thus rising or falling relative to the level of the final pitch accent, associated with the nucleus. My concern is with this *nuclear tone*, as it is traditionally called in the British school of intonational studies, or *nuclear tune*, as I will refer to it here (following Gussenhoven 2001) — that is, the part of the overall tune comprised by the nuclear accent plus the pitch movements that follow it, up to the terminus. For all the cases I consider, the terminus of the utterance will coincide with the end of a sentence.

Following Gussenhoven 1983 (and its precedents in the British tradition), and consistent in spirit with the approach of Ladd (1980, 1983),¹ I will assume three central categories of nuclear tunes:

- Rise
- Fall
- Fall-rise

Of these I treat only the categories of the rise and the fall in this thesis, ignoring the complexities of the fall-rise. Instances of the rise and fall are roughly characterizable according to their F₀ (fundamental frequency) shape as follows:

- (11) Rise: Non-falling from the nuclear pitch accent to the terminus and ending at a point higher than the level of the nuclear accent.
- (12) Fall: Non-rising from the nuclear pitch accent to the terminus and ending at a point lower than the level of the nuclear accent.

Rising and falling nuclear tunes are simple in the sense that they are unidirectional. A fall-rise, by contrast, as its name suggests, falls from the pitch accent and subsequently rises to the terminus. The greater complexity in shape of the fall-rise is mirrored by complexities in its distribution. On declaratives, it does not have as free a distribution as a simple falling tune; the restrictions on distribution have been associated with its interpretation as a ‘topic’ marker, a term used by Jackendoff 1972. Furthermore, in comparison to simple rises, fall-rises on polar interrogatives are restricted in distribution, which complicates the minimal-

¹ The system of Ladd 1980 has much in common with Gussenhoven’s (see Ladd 1996:290, fn. 8), and I believe the assumptions I make here to be largely compatible with the claims of that work as well. However, Ladd 1980 proposes two distinct categories of rises, the low-rise and the high-rise, whereas I will assume these are subtypes within a single category. In later work (Ladd 1983, fn. 13) Ladd suggests representing both types of

pair strategy pursued here. For simplicity I therefore exclude the category of the fall-rise from the present study, together with any other nuclear tunes that do not meet the descriptions in (11)-(12). (Note that the categories of the rise and fall as characterized in (11)-(12) do not correspond exactly to Gussenhoven's, although they are quite close.)

In referring to the rise and fall as intonational *categories*, I intend to emphasize my assumption that the terms 'rising' and 'falling' each characterize a family of contours – a natural class – rather than being identified with or limited to a single tune within that class. This assumption does not preclude the use of a tonal transcription system. In Gussenhoven's system, the three categories *rise*, *fall*, and *fall-rise* are associated with the autosegmental representations ★LH, ★HL, and ★HLH, respectively. The categories described by (11)-(12) can, however, be described in other systems as well, even systems that do not recognize rises and falls as intonational entities. For example, the above description of the rise fits all of the tunes H* H H%, L* H H%, L* H L%, and L* L H% in the system of Pierrehumbert 1980, as modified in Beckman and Pierrehumbert 1986. Observe that there is no single element – pitch accent, phrase tone, or boundary tone – that is common to these representations. In particular, the H% boundary tone, which would seem to be a natural candidate for characterizing a final rise, is not present in L* H- L%, which is nevertheless 'rising' in the descriptive sense of (11). That is

rise as LH, distinguishing the two with a feature; that revision is compatible with my assumptions and Gussenhoven's categories.

the main practical reason why I refrain from identifying the rise with a particular tonal element in that system.²

In Gussenhoven's system, variations within each of the three basic categories are handled in terms of a set of systematic 'modifications' that apply in a consistent way across the categorial types. (The idea is similar in general outline to Ladd's featural specifications; see Ladd 1980, 1983.) These modifications influence the meaning conveyed in each case, so that variants within a category may have different nuances while having in common a semantic core associated with the category. Following Gussenhoven, I assume a taxonomy of this sort. However, I ignore any possible contributions of the modificational elements, investigating only the major categorial contrast between the rise and the fall.

What these assumptions amount to in practical terms is that the reader is free to use any simple rising contour for the (a) and (b) sentences in any particular example, as long as the contour is held constant in the comparison of those particular interrogative and rising declarative sentences. Similarly, the falling intonation assumed for any particular instantiation of the (c) examples can be any contour qualifying under (12). This does not lead to the expectation that all contours within a category are freely interchangeable across all contexts. As just

² On a more theoretical note, I point out that the H vs. L distinctions encoded in the Beckman/Pierrehumbert system are mediated through a system of phonological rules, resulting in a certain amount of abstraction in the representations, such as the use of H- L% to represent a high plateau. However well motivated the representations are for the purposes of capturing phonologically significant distinctions between contours with a minimum of machinery, it does not follow that a particular combination of tonal values and rule mechanisms chosen largely on grounds of conciseness and efficiency in phonological terms is going to map transparently to a system of *semantic* distinctions. See Pierrehumbert 1980, Chapter 1, for comments on the relatively minor role assigned to semantic factors in the design of the original system.

discussed, Gussenhoven's system allows for the possibility that there are nuanced differences between members of a category, and these differences may interact with contextual factors to make some possibilities less natural than others in a given case.

If this freedom of choice proves onerous, a compromise can be reached by choosing a single contour of each type to use throughout the thesis. For rising intonation, I recommend as the most versatile of rising tunes the contour known as the 'high-rise' ($H^* H H\%$ in the Pierrehumbert system). For falls, $H^* L L\%$ (the so-called 'declarative fall') can be used. In any event, my working assumption about the exact membership of the categories of 'rising' vs. 'falling' intonation can be distinguished, at least in principle, from the hypothesis about the meanings associated with those categories that constitutes a major claim of the thesis. Readers who disagree with the phonological assumptions outlined here are invited to substitute their own categories for the distinctions marked by '?' and '.' throughout the thesis.

In the minimal-pair comparison, all factors other than the elements being compared are to be held constant. This includes the location of the nuclear accent and any pitch accents other than the nuclear accent. The location of the nuclear accent, as is well known, affects (or is a reflex of) the focus-background structure of a sentence. That is true whether the sentence is declarative or interrogative, rising or falling. To avoid this orthogonal complication, the examples throughout this thesis are to be read (wherever possible) with the nuclear accent placement

associated with broad, ‘all-new’ focus. Issues related to the notion of ‘topic’ are also orthogonal to the present study, since I exclude from consideration the category of nuclear tune usually associated with topic in English, the fall-rise.

One final note. Many – though not all – of the examples I cite are based on naturally occurring speech, my own as well as that of others. It is in the nature of the minimal-pair strategy, however, that judgments must be made about non-occurring data as well as the sentences actually attested. The judgments throughout are my own, and the reader may want to know that I am a native English speaker, born and raised in Seattle. I mention this for completeness. With respect to the categories of simple rises and falls, and the category of polar questions considered here, I do not know of significant differences with other dialects of English.

1.3 Previous accounts

The task of locating the present account in the context of previous work about intonation and sentence type is very difficult in one way and quite easy in another. It is difficult in the sense that there is a great deal of work in many of the areas that this thesis touches upon. In particular, there is a vast literature on intonational structure and meaning, replete with acute observations and proposals. But the task is easy in the sense that very little of this literature treats the subject of this thesis, the interaction of intonation with sentence type. Even though authors often pay

attention to sentence type in a general way, explicit hypotheses about the interaction of intonational meaning with sentence type are rare.

Focusing in on declarative sentence type and the use of declaratives as questions, the field becomes narrower still. Declaratives as a semantic category have received relatively little attention, although accounts of speech acts (such as assertion) that can be performed with falling declaratives abound. Analyses that do associate a speech-act meaning with declarative sentences, such as Ross 1970 or Katz 1977, are implicitly restricted to declaratives with falling intonation and their uses. Formal studies of questioning, on the other hand, generally concentrate on the interpretation of syntactic interrogatives and their canonical use as requests for information.

There are, however, two recent lines of work that bear closely on the specific concerns of this thesis. Bartels, in her 1997 dissertation, analyzes how the choice of intonational tune contributes to the functioning of a sentence as a statement or question. She focuses in particular on the latter category, defined in functional terms. The contribution of sentence type is not expressed formally but does figure in the analysis in a systematic way, which I will comment on in Chapter 5. This dissertation thus follows Bartels's lead in concentrating on the intersection of tune, sentence type, and a pragmatic category of 'questioning', although different conclusions are reached in each case. We share, in addition, a number of fundamental assumptions about the nature of intonational meaning, as well as the reliance on minimal-pair methodology. Despite the overlap in terrain, the two

studies are rather different in emphasis and are largely complementary. Whereas Bartels takes a panoramic view, including *wh*- and alternative questions in her survey as well as standard polar interrogatives and the full range of intonational contours, I concentrate on providing a formal and explicitly compositional account for a subset of sentence types and contours.

From a different perspective, Beun explores the properties of Dutch declarative questions in series of articles (1989, 1990, 1994, 2000). Beun's work is based on a corpus of elicited dialogues involving an information clerk at an airport and information-seekers with the task of making travel plans. The focus of Beun's investigations is on how discourse participants recognize declaratives as questions and what factors correlate with the choice of declarative form to express a question. Unlike Bartels 1997 and the present study, Beun is not particularly interested in intonational meaning. Furthermore, his results are for Dutch, not English. Nevertheless, Beun's observations about the contexts in which declaratives are interpreted as questions are quite relevant to this thesis, particularly with respect to the hypothesis about declarative questioning defended in Chapter 4, with which they are largely compatible. I will comment on Beun's findings in that discussion.

The idea that rising vs. falling intonation is related to a Speaker/Addressee distinction has precedents in the intonational literature, though none are developed in the particular direction taken here. In recent work, Steedman 2000 proposes that the H% vs. L% boundary tone distinction (using the Beckman and Pierrehumbert system) correlates with 'ownership' of the content expressed. The proposal of

Merin and Bartels 1997 that the rise ‘alienates choice to Alter’ while the fall ‘appropriates choice for Ego’ offers a related idea as well. Noh’s 1998 Relevance-Theoretic discussion of ‘echo questions’ and their kin relies on the idea of attributing the thought expressed (or a related thought) to the Addressee, although Noh does not single out intonation as a factor. The present account is compatible in a broad way with these suggestions and can be seen as a development of the shared core notion of tying an intonational contrast to a Speaker/Addressee distinction.

Chapter 2

The Distribution of Declarative Questions

2.1 *Introduction*

This chapter documents and discusses two major generalizations concerning the distribution of declaratives as questions. First, in Section 2.2, I show that there are many contexts that allow interrogatives but exclude both rising and falling declaratives as questions. The hypothesis extended to cover these cases is that declaratives, unlike interrogatives, express bias – they cannot be neutral.

Section 2.3, on the other hand, will show an important difference between rising and falling declaratives, documenting ways in which rising declaratives form a class with interrogatives. The generalization illustrated in this section is that neither rising declaratives nor interrogatives express commitment on the part of the Speaker. There is thus considerable flexibility in the attitude attributed to the Speaker in a particular context, and in the use to which a rising declarative or interrogative question may be put. It will be seen that falling declaratives are less flexible in this respect and *do* commit the Speaker to their content.

The empirical study in this chapter can be seen as providing support for a natural set of intuitions about rising intonation and declaratives: the rise is associated with lack of commitment (on the Speaker's part), and declarative form

has an element of bias, or what we may loosely call ‘assertiveness’. But as Section 2.4 points out, there is tension between these two ideas, particularly in the case of rising declaratives. The challenge is to understand how a rising declarative can simultaneously result in bias and in lack of commitment, and to do this in a way that leads to testable predictions. The chapter closes with a sketch of the solution adopted here, and a preview of its implementation in Chapter 3.

A third major generalization concerning the distribution of declarative questions, the Bias Condition, is reserved for discussion in Chapter 4.

2.2 *Declarative bias*

In this section I document ways in which rising declaratives pattern with their falling declarative counterparts, differing from interrogatives. The central observation is that declaratives are unsuitable in contexts where the Speaker is expected to maintain an attitude of neutrality or ignorance. First, as noted in the introduction, rising declaratives do not work to elicit information in an unbiased way, as (13)-(15) show.

(13) [on a tax form]

- a. During the tax year, did you receive a distribution from a foreign trust?
- b. #During the tax year, you received a distribution from a foreign trust?
- c. #During the tax year, you received a distribution from a foreign trust.

(14) [in a guessing game]

- a. Is it bigger than a breadbox?
- b. #It’s bigger than a breadbox?
- c. #It’s bigger than a breadbox.

- (15) [as an exam question]
- a. Is the empty set a member of itself?
 - b. #The empty set is a member of itself?
 - c. #The empty set is a member of itself.

In a similar vein, (16)-(17) show that the issue raised by a declarative question cannot be regarded as open or unsettled, liable to go either way. In fact, in (18) the rising declarative cannot be described as a *question* at all, even though the construction is one that accepts a root clause, as (18a) demonstrates.

- (16) It's an open question.
- a. Did she lie to the grand jury?
 - b. #She lied to the grand jury?
 - c. #She lied to the grand jury.
- (17) a. Will the incumbent win re-election? It could go either way.
 b. #The incumbent will win re-election? It could go either way.
 c. #The incumbent will win re-election. It could go either way.
- (18) a. The question is, does he have the money?
 b. The question is, #he has the money?
 c. The question is, #he has the money.

Interrogatives can be used to initiate a line of inquiry and hypothetically extend it using *if so* and *if not*, as seen in (19)-(20). For example, (19a) is the sort of question that might appear on a health insurance form. Rising declarative questions cannot be used in this way.

- (19) Are you married?
- a. If so, does your spouse have health insurance?
 - b. #If so, your spouse has health insurance?
 - c. #If so, your spouse has health insurance.
- (20) Does Gene own a cell phone?
- a. If not, would he like to buy one?
 - b. #If not, he'd like to buy one?
 - c. #If not, he'd like to buy one.

(21)-(22) demonstrate that rising declaratives, unlike interrogatives, don't work well to solicit advice or an opinion – what Huddleston 1994 calls 'direction' questions.

- (21) What do you think?
 - a. Has the stock market bottomed out?
 - b. #The stock market has bottomed out?
 - c. #The stock market has bottomed out.
- (22) What do you think?
 - a. Should I cut my hair?
 - b. #I should cut my hair?
 - c. #I should cut my hair.

The same point holds for self-addressed deliberative questions:

- (23) Dieter contemplating a tray of pastries:
 - a. Am I (really) hungry? Do I need this chocolate doughnut?
 - b. #I'm really hungry? #I need this chocolate doughnut?
 - c. #I'm really hungry. #I need this chocolate doughnut.
- (24) Homeowner glancing out the window:
 - a. Let me see, does the lawn need mowing?
 - b. Let me see, #the lawn needs mowing?
 - c. Let me see, #the lawn needs mowing.

Rising declaratives make poor speculative questions, i.e., questions designed to instigate thought and/or discussion without necessarily being answered or answerable. (26a), for example, might lead into a discussion of the JFK assassination without committing the Speaker to any particular view; (26b) cannot be used for the same effect.

- (25) a. Does God exist?
 - b. #God exists?
 - c. #God exists.

- (26) a. Did Oswald act alone?
b. #Oswald acted alone?
c. #Oswald acted alone.

As is already evident, the patterns involving restrictions on declaratives as questions are not limited to standard ‘information question’ contexts, i.e., requests for information from an uninformed Speaker to an Addressee assumed to be informed and willing to provide the information. The point becomes even clearer when we look at examples like (27)-(28), in which interrogatives function as polite requests for action rather than for information. Declaratives do not share this function, as the (b) and (c) cases show.

- (27) a. Can you (please) pass the salt?
b. #You can (please) pass the salt?
c. #You can (please) pass the salt.

- (28) a. Would you (please) sit down?
b. #You would (please) sit down?
c. #You would (please) sit down.

The sarcastic questions in (29)-(30) provide another illustration of the reduced rhetorical range of declarative questions compared to interrogatives.

- (29) a. Is the Pope Catholic?
b. #The Pope’s Catholic?
c. #The Pope’s Catholic.

- (30) a. Do bears shit in the woods?
b. #Bears shit in the woods?
c. #Bears shit in the woods.

The descriptive generalization I advance for the examples considered so far is given in (8):

- (8) Declaratives express a bias that is absent with the use of interrogatives; they cannot be used as neutral questions.

In offering (8) as a descriptive generalization for the data so far I also offer an implicit hypothesis about the use of interrogatives in the contexts illustrated — namely, that certain functions of interrogatives, such as direction questions, polite requests, etc., involve at least the appearance of neutrality. I won't attempt to justify this hypothesis explicitly, which would require case-by-case study of the various uses seen above. Rather, I will take (8) as a reasonable working descriptive generalization and seek a characterization of the notions of neutrality and bias, with the expectation that such notions will ultimately be useful in understanding the range of discourse functions available for interrogatives as well as declaratives. It should also be noted that *negative* polar interrogatives behave quite differently from their positive counterparts and cannot be freely substituted for them; examples will be given below.

The flip side to the patterns seen so far is that declaratives, differing from interrogatives, are useful in situations where bias rather than neutrality is called for. When it comes to contributing new information, for example, bias is a good thing. This is a given for falling declaratives, which are the prototypical way to offer a piece of news. But rising declaratives, too, have this potential for many speakers. The use of rising declaratives as a routine way to offer new information is exemplified in (31)-(32). (A similar example is cited by Pierrehumbert 1980 in

discussing the difficulties of isolating and characterizing the contribution of intonational meaning.) Note that rising interrogatives do not share this function.

- (31) Radio station DJ: Good morning Susan. Where are you calling from?
Caller:
a. #Am I from Skokie?
b. I'm from Skokie?
c. I'm from Skokie. [adapted from Hirschberg and Ward 1995]
- (32) a. #Is my name Carl? #Will I be your waiter tonight?
b. My name is Carl? I'll be your waiter tonight?
c. My name is Carl. I'll be your waiter tonight.

The main concern of this paper is the use of rising declaratives as questions; but the possibility of informative use must be allowed for by the analysis.

Huddleston 1994 observes that interrogatives are incompatible with certain 'bias markers', among them *of course*, *no doubt*, and *surely*:

- (33) a. #Has the manager *of course* been informed?
b. The manager has *of course* been informed?
c. The manager has *of course* been informed.
- (34) a. #Did they *no doubt* misunderstand her intentions?
b. They *no doubt* misunderstood her intentions?
c. They *no doubt* misunderstood her intentions.
- (35) a. #Are you *surely* going to agree?
b. You're *surely* going to agree?
c. You're *surely* going to agree.

Similar results obtain for evidential adverbs, shown in (36), and for *therefore*, as in

- (37):
- (36) a. #Has he evidently/apparently left already?
b. He's evidently/apparently left already?
c. He's evidently/apparently left already.

- (37) A: Laura's car is in the parking lot.
B's response:
a. #Therefore is she here?
b. Therefore she's here?
c. Therefore she's here.

The examples given throughout this section distinguishing the use of declaratives from that of interrogatives tally with other systematic differences between the two categories. For example, interrogatives, but not declaratives, support polarity items like *any* and *ever* (Hirst 1983, Huddleston 1994):

- (38) a. Is anybody home?
b. #Anybody's home?
c. #Anybody's home.
- (39) a. Did he ever finish?
b. #He ever finished?
c. #He ever finished.

More generally, rising declaratives behave like declaratives with respect to negation and polarity items, and not like interrogatives. The negative declaratives in (40b-c), for example, are straightforwardly understood as expressing the proposition that the doctor is *not* in; their bias is accordingly negative. What exactly is conveyed by the negative interrogative in (40a) is much harder to characterize, and the bias is not necessarily negative. (In fact, negative polar interrogatives are systematically ambiguous, as Ladd 1981 shows.)

- (40) a. Isn't the doctor in?
b. The doctor isn't in?
c. The doctor isn't in.

The point is that the negative rising declarative does not present the same complexities as the interrogative version. I restrict attention to positive

interrogatives in this paper, as they clearly display the potential for neutrality that is at issue in the contrast with declaratives.

Furthermore, rising declarative questions cannot be made into polar alternative questions with *or not*, as interrogatives can be:

- (41) a. Did she order coffee or not?
b. #She ordered coffee or not?
c. #She ordered coffee or not.

- (42) a. Is it raining or not?
b. #It's raining or not?
c. #It's raining or not.

In fact, the point is more general. Rising declarative questions do not make good alternative questions of any sort. The interrogative in (43a) can be read in two ways, depending on intonation and phrasing: on one reading it asks which of the two beverages was ordered, while on the second the question is whether a beverage was ordered at all. The rising declarative in (43b), as a question, has only the second kind of reading; the falling declarative is also unambiguous (though as usual for falling declaratives, it does not easily receive a question reading).

- (43) a. Did she order coffee or tea?
b. She ordered coffee or tea?
c. She ordered coffee or tea.

Alternative questions thus provide another example of a possibility available only with true syntactic interrogatives, not declaratives, regardless of intonation.

The facts about sentence type, polarity marking, and alternative questions exemplified in (38)-(43) do not fit in any obvious way under the generalization in (8), and I will have nothing further to say about them in this dissertation. But in

providing another instance of how declaratives pattern together, they do support the general view defended here – namely, that rising declarative questions are not interchangeable with rising interrogatives, but acquire their questioning function in a more complex way that is faithful to their essentially declarative nature.

2.3 *Lack of Speaker commitment*

A second crucial observation about the distribution of rising declaratives is that they are far more natural as questions than their falling declarative counterparts. In this section I support that intuition empirically by showing that rising declaratives pattern in certain ways with the corresponding rising interrogatives, differing from falling declaratives. The generalization advanced is that rising declaratives, like interrogatives, fail to commit the Speaker to their propositional content. This point will emerge in comparisons with falling declaratives, which evidently do commit the Speaker.

First note that rising declaratives, like interrogatives, allow for readings in which the Speaker is understood to be skeptical of the proposition expressed. In (44), either an interrogative or a declarative may be used to acknowledge and/or elliptically reiterate A's utterance; but only (a) and (b) are compatible with B's follow-up remark, which implies that B remains doubtful about the alleged improvement. The falling declarative in (44c) seems to express overt agreement

with A's opinion, and thus has the effect of inconsistency with the skeptical follow-up.

(44) [A&B are looking at a co-worker's much-dented car]

A: His driving has gotten a lot better.

B's response:

- a. Has it? I don't see much evidence of that.
- b. It has? I don't see much evidence of that.
- c. It has. #I don't see much evidence of that.

The skeptical reading of rising declaratives is well-known, and is often connected with their "echoing" function. But it would be a mistake to assume that rising declaratives are inherently skeptical (or inherently echoing, for that matter). Rising declaratives, like interrogatives, also allow for readings in which the Speaker is understood as routinely accepting the proposition expressed, as illustrated in (45)- (46). Falling declaratives are acceptable in these cases as well.

(45) A: That copier is broken.

B's response:

- a. Is it? Thanks, I'll use a different one.
- b. It is? Thanks, I'll use a different one.
- c. (Oh), it is. Thanks, I'll use a different one.

(46) A: Jake's here.

B's response:

- a. Is Jake here? Then let's get started.
- b. Jake's here? Then let's get started.
- c. (Oh), Jake's here. Then let's get started.

An example that does not involve echoing (in a strict sense, at least) is given in (47). Here the question raised concerns a presupposition of A's utterance, rather than the main proposition expressed. Again, both the interrogative and the rising declarative are fine, while the falling declarative is unacceptable. (We will see other

examples of acceptable non-echoing uses of rising declaratives, along with limitations on such uses, in Section 4.2.)

- (47) A: The king of France is bald.
B's response:
a. Is France a monarchy?
b. France is a monarchy?
c. #France is a monarchy.

Like the earlier examples, (47a-b) are compatible with either skepticism (*France is a monarchy? Since when?*) or acceptance (*France is a monarchy? I didn't realize that.*) by the Speaker. It is the follow-up remark that provides the clue to the Speaker's attitude. (47a-b) by themselves are noncommittal, imposing no constraints on interpretation of the Speaker's position. What all such readings have in common, however, is the sense that the proposition expressed is newsworthy, from the Speaker's point of view – something not known before.

Rising declaratives, like interrogatives, may be used to make the point that the Addressee, rather than the Speaker, is in a position to know whether the proposition expressed is true. Consider (48)-(49):

- (48) a. Is shoplifting fun?
b. Shoplifting's fun?
c. #Shoplifting's fun. [# as an attempt to insinuate that the Addressee has shoplifted]
- (49) a. Was the food good in jail?
b. The food was good in jail?
c. #The food was good in jail. [# as an attempt to convey that the Addressee has been in jail]

To appreciate what sets apart (48a-b)-(49a-b) from (48c)-(49c), suppose that (48a) is uttered in circumstances like the following. The Speaker, in a recent visit to the

local mall, has observed the Addressee being arrested for shoplifting. The Addressee is not aware of the Speaker's knowledge; it is therefore not mutual knowledge. When the two parties meet, the Speaker can ask not-so-innocently: *Is shoplifting fun?* This has the effect of communicating that the Addressee is known to be a shoplifter. The answer given by the Addressee is immaterial; the damage is done by the question itself. What this seems to indicate is that use of an interrogative carries an assumption on the level of a presupposition that the person to whom a question is addressed is knowledgeable on the subject. (Hudson (1975:12) proposes a semantic condition on polar interrogatives of exactly this sort, involving the relative knowledgeability of Speaker and Addressee.) What makes the insinuation work so well in the above examples is the presence of a predicate like *fun* or *good*, which calls for subjective evaluation. Compare *Is shoplifting a crime?*, which does not have the same effect. It generally takes firsthand experience to judge whether an activity is fun or not. Thus, if the Addressee is assumed to be knowledgeable about whether shoplifting is fun, it will also be assumed (in the usual case) that the Addressee has shoplifted.

Consistent with the claim of this section that rising interrogatives fail to commit the Speaker, the Speaker who asks (48a) does not incriminate herself in any way. By comparison, the most natural readings of the falling declaratives in (48c)-(49c) portray the Speaker as the source of information, suggesting that it is the Speaker who has shoplifted and/or been jailed.

The (b) cases of (48) and (49) require context of a particular sort to be understood as questions; rising declarative questions cannot be used out of the blue in the same way that the interrogative versions can be, a restriction to be addressed in Chapter 4. The important point at present, though, is that when understood as questions, the rising declaratives in (48b)- (49b) pattern with the (a) examples in not conveying anything about the *Speaker's* criminal habits. This again supports the hypothesis that both rising declaratives and interrogatives fail to commit the Speaker to their propositional content.

The twin generalizations that emerge from the observations in this section are given in (50)-(51):

(50) Rising declaratives, like interrogatives, fail to commit the Speaker to their content.

(51) Falling declaratives do commit the Speaker to their propositional content.

It follows from (50) that rising declaratives, like interrogatives, allow for a range of Speaker attitudes to be attributed. Falling declaratives, on the other hand, are compatible only with attitudes consistent with commitment.

In the next section I will take up the problem of how the lack of commitment illustrated for rising declaratives in this section can be squared with the observations about declarative bias given in Section 2.2. Before moving on, however, there is an additional point to be made about the data in this section. It is noteworthy that in all of the above examples, the acceptability of a rising declarative as a question is paralleled by the acceptability of the corresponding

interrogative. Furthermore, the interrogative and rising declarative versions are intuitively very similar in effect, whatever that effect happens to be in a particular case. This is especially striking since the “questions” exemplified in (44)-(49) are atypical in at least one important respect – they ask a question the answer to which has already been given or implied by the Addressee. To find rising declaratives used in this way is nothing new, since their function as echo questions is well known (though less well understood). But the fact that an interrogative can be used in the same context to much the same effect is significant and, as I will argue in Section 4.3, provides an important clue to the puzzle of rising declarative questions. For now I ask the reader simply to take note of the phenomenon.

2.4 Reconciling bias with lack of commitment

The data and descriptive generalizations advanced so far seem to lead naturally to a compositional understanding of rising declaratives in terms of their defining components, the rise and declarative form. The outline suggested by the data is as follows:

Understanding the bias

- (52) The intuition: A declarative question expresses (some degree of) commitment to the proposition expressed, consistent with interpretation of the sentence as having an element of ‘assertiveness’ or ‘bias’. (See, e.g., Bolinger (1957), Huddleston (1994).)

Understanding the questioning use

- (53) The intuition: The rise expresses lack of commitment to the proposition expressed, consistent with interpretation as a question.

Equally intuitively, these two generalizations seem contradictory as stated. The challenge to be addressed in this section is how to reconcile them, and thus integrate the explanations of the observations in Sections 2.2-2.3, in a principled way.

The hypothesis I will implement is given informally in (54)-(55):

- (54) Rising declaratives commit the Addressee to the proposition expressed.
- (55) Falling declaratives commit the Speaker to the proposition expressed.

In compositional terms, I will take declaratives to express *commitment* to their propositional content p by some discourse participant, where commitment is understood as ruling out the alternative, $\neg p$. The intonational component will specify which participant is committed: the Addressee in the case of the rise, the Speaker in the case of a fall.

The above proposal resolves the tension between lack of commitment and bias in the following way. Rising declaratives do fail to commit the *Speaker* to p , as (53) suggests. But rising declaratives do express commitment to p on the part of the *Addressee*, allowing for an understanding of the bias.

In the account to follow in Chapter 3, bias will be characterized in contextual terms. The use of either a rising or a falling declarative expressing p ensures that the context is one in which the participants cannot easily come to agreement on $\neg p$. By committing one participant to p , the declarative rules out $\neg p$ as a mutual assumption, effectively conveying a bias toward p . The contrast with interrogatives

also follows in a natural way: I propose that interrogatives, unlike declaratives, commit nobody to their content and thus have the capacity to be neutral.

Chapter 3

Modeling Bias and Neutrality

3.1 *The discourse context*

The proposal sketched in Section 2.4 specifies that a falling declarative commits the Speaker to its content, while a rising declarative commits the Addressee. To implement this proposal, I begin with the familiar notion of the Common Ground (Stalnaker 1978). Under Stalnaker's classic treatment, the Common Ground (hereafter CG) is a set of propositions representing what the participants in a discourse take to be mutually believed, or at least mutually assumed for the purposes of the discourse. I assume, as Stalnaker does, a framework in which a proposition is construed as a set of worlds, the worlds of which the proposition is true. The CG can be defined in propositional terms as follows:

- (56) $CG \text{ of a discourse} = \{ p \in \wp(W) : p \text{ is a mutual belief of the participants in the discourse} \}$

Equivalently, the context can be treated as a set of worlds, $\cap CG$, the worlds of which all of the propositions representing mutually held beliefs of the participants are true. In Stalnaker's terminology, this set of worlds is the *context set* of the discourse.

(57) context set of a discourse = { $w \in W$: the mutual beliefs of the discourse participants are true of w }

The mutual beliefs constituting the CG are, crucially, *mutual* and not just shared.

That is, for each p in the CG, each participant is not only taken to believe p but to be aware that other participants believe p as well; and they, likewise, not only share the belief but recognize that the belief is shared. In other words, mutuality involves beliefs about one's own and others' beliefs.³ Beliefs that remain private, and beliefs that the participants happen to have in common without mutually realizing it, are not part of the CG.

Although the CG is typically employed, as it will be here, to theorize about linguistic contributions to the discourse, it is important to remember that it ordinarily contains much more than the propositional content of statements. Propositions contributed by implicatures of what is said are also in the CG. And beyond the mutual inferences formed in response to speech events, the participants come to the discourse, even if they are strangers to each other, with mutual beliefs of a general sort, including the expectation of their use of a shared language, assumptions relating to their membership in a particular culture and speech community, and mutual observations relating to the physical environment of the discourse (time, place, communicative medium, etc.) If the participants know each other, slightly or well, they will also have mutual beliefs relating to their personal

³ How to characterize mutual knowledge, or belief, in a way that captures its recursive nature and yet is cognitively plausible is a matter of some debate, which I will not enter into. For present purposes I simply assume that the kind of mutual belief context required for communication is characterizable in some way consistent with the proposal outlined here.

histories and previous interactions. In addition to these pre-existing beliefs, a multitude of mutual beliefs will be formed about salient events occurring as the discourse progresses. These salient events predictably include, at a minimum, the speech events that make up the discourse proper; the participants can be expected to form mutual beliefs about the *facts* of these events – that they took place at a given time or in a certain sequence, which participant uttered what, etc. More exotic events, such as a goat unexpectedly entering the room where the discourse is taking place, will also result in additions to the CG.

The content of the CG thus depends on who the discourse participants are, as well as the circumstances of the discourse. Indeed, the participants must be mentioned, as they are in (56)-(57), to give any sort of characterization of the context in terms of mutual beliefs or assumptions. (The related notion of a *conversational background* (Kratzer 1981) also makes reference to a group of individuals, the speech community.) As an indicator of this dependency, I will use the notation $CG_{\{A,B\}}$ for a CG in which A and B are the discourse participants. I assume two discourse participants throughout.

The step needed to implement the hypothesis in (54)-(55) in a straightforward way is to separate out the beliefs publicly attributed to each participant, as is done in (58).

- (58) Let $CG_{\{A,B\}}$ be the Common Ground of a discourse in which A and B are the individual discourse participants.
- a. DC_A of $CG_{\{A,B\}} = \{ p: 'A \text{ believes } p' \in CG_{\{A,B\}} \}$
 - b. DC_B of $CG_{\{A,B\}} = \{ p: 'B \text{ believes } p' \in CG_{\{A,B\}} \}$

(58) defines a more articulated version of the CG, without making any essential changes to the conception. The set of propositions associated with each participant represents what we may think of as their *public beliefs*, or *discourse commitments* (DC) – public in the sense that the participant is mutually recognized as committed to them. All mutual beliefs are public in this sense, but a public belief of an individual does not have to be mutual; an illustration will be given below.

The definition in (58) takes $CG_{\{A,B\}}$ as the basic structure and defines DC_A and DC_B in terms of $CG_{\{A,B\}}$. But we can just as easily take DC_A and DC_B as basic, deriving $CG_{\{A,B\}}$. And since I am concerned here with public beliefs of individual participants and not just mutual beliefs, the latter orientation, as spelled out in (59), will be more convenient.

- (59) Let DC_A and DC_B be sets of propositions representing the public beliefs of A and B, respectively, with respect to a discourse in which A and B are the participants, where:
- a. p is a public belief of A iff ‘A believes p ’ is a mutual belief of A and B
 - b. p is a public belief of B iff ‘B believes p ’ is a mutual belief of A and B

In light of (59), the context of the discourse can be represented as an ordered pair $\langle DC_A, DC_B \rangle$, replacing $CG_{\{A,B\}}$ (which is still derivable as $\{ p: p \in DC_A \ \& \ p \in DC_B \}$.) Or, equivalently and more conveniently, the context can be construed as an ordered pair of sets of worlds, analogous to Stalnaker’s notion of the context set. I adopt the latter construal as the representation of the context, as shown in (60). Here the abbreviation *cs* stands for the *commitment set* of an individual, the set of worlds of which that individual’s public beliefs are true.

- (60) Let a discourse context $C_{\{A,B\}}$ be $\langle cs_A, cs_B \rangle$, where:
 A and B are the discourse participants
 a. cs_A of $C_{\{A,B\}} = \{ w \in W : \text{the propositions representing A's public beliefs are all true of } w \}$
 b. cs_B of $C_{\{A,B\}} = \{ w \in W : \text{the propositions representing B's public beliefs are all true of } w \}$

Just as the original Stalnakerian concept of the Common Ground is recoverable given DC_A and DC_B , the context set is recoverable from $\langle cs_A, cs_B \rangle$: it is the set of worlds of which all *mutual* beliefs of A and B are true, namely $cs_A \cup cs_B$.

If a participant has no public beliefs, there is no restriction on the set of worlds compatible with her public beliefs; the individual commitment set in such a case will be W , the set of all worlds. In practice this is not a realistic possibility, given that participants start out with some mutual beliefs even before any speech acts take place, as the discussion above pointed out. Even if the discourse proceeds in such a way that a participant retracts or revises a public belief, possibly expanding the commitment set as a result, we do not normally expect a commitment set to revert to W . In fact a major difficulty in modeling revision (or a transition to counterfactuality) is specifying which proper subset of W is the appropriate one to arrive at. Nevertheless, there is no reason to rule out the possibility in principle of having W as a commitment set.⁴

The case of an empty commitment set, on the other hand, would represent a situation where the participant's public beliefs are inconsistent – there is no world

⁴ In discourses involving counterfactual assumptions or, more generally, non-epistemic assumptions, the discourse commitments of participants will differ in at least some respects from their epistemic commitments. In such situations it becomes more reasonable to imagine a participant starting out with a *discourse*

such that all of the beliefs can be true of it. In practice I assume participants do make inconsistent moves, some of which they recognize as inconsistent, and others which they don't. About the latter category I have nothing to say here (inconsistent moves not recognized as such), and I will, following standard practice, ignore them. About the former category (inconsistent moves recognized as such), I will assume that while discourse participants may make moves that apparently result in inconsistency and thus an empty commitment set, they do not proceed with the discourse in such a state. To the extent they recognize the inconsistency, participants will take it as an indicator that the context is in need of adjustment, rather than continuing to attribute inconsistent commitments to themselves or each other. Reaching an empty commitment set, then, is assumed to trigger a repair operation by which participants salvage what they can to retain a non-empty commitment set. I will have nothing to say about the salvage methods involved. The contextual states and operations I am concerned with will only be defined for contexts in which the necessary adjustments have already been made. To that end I define empty commitment sets and contexts in (61)-(62):

(61) $cs_X(C)$ is *empty* iff $cs_X(C) = \emptyset$.

(62) C is *empty* iff there is at least one empty cs_X in C .

I will generally use the notation $cs_X(C)$, as seen above, to refer to the commitment set of some discourse participant X in the context C . For notational ease, I will normally drop the subscript of $C_{\{A,B\}}$, as above; references to C , as in (61)-(62), are

commitment set of W , whatever the mutual *epistemic* commitments are at the outset. I will not consider these

to be understood as references to contexts with two participants, usually identified as A and B.

As an illustration of what is accomplished by the divided context as defined in (60), consider a discourse in which A and B publicly disagree on some point. Suppose, for example, that A has said that cats make better pets than dogs, while B has argued in favor of dogs. Let q stand for the proposition expressed by *Cats make better pets than dogs*. Clearly q is not a mutual belief of A and B, since A and B are in disagreement on this point. Of course $W-q$ (the set of worlds of which $\neg q$ is true) is not a mutually held proposition either, since A's belief is in conflict with it. Still, q does figure indirectly in the Common Ground, which still records their mutual beliefs about each other's positions. That is, the fact that A believes q itself has the status of a mutual belief, as does the fact that B believes a proposition entailing $W-q$ (assuming, as I do here, that the participants' statements can be taken as an index to their beliefs). The situation differs crucially from one in which the opinions of A and B on the matter remain private, and this difference is reflected in the representation of the context.

The formalism just introduced makes the descriptive task easy in situations like the one described above: we can say that q is a discourse commitment (a public belief) of A's (that is, $cs_A \subseteq q$), and $W-q$ is a discourse commitment of B's ($cs_B \subseteq W-q$). Clearly neither q nor $W-q$ can become a mutual belief of A and B in this context, at least not without one participant revising their position. Let us call this

sort of situation one in which both q and $W-q$ are *controversial* with respect to the context. (A formal definition will be given below.) The relevant notions are characterized in (63)-(66) below.

Status of a proposition p with respect to a discourse context C :

- (63) p is a *commitment* in C of an individual participant X iff cs_X is not empty and $cs_X \subseteq p$.
- (64) p is a *joint commitment* in C iff both discourse participants are committed to p .
- (65) p is *resolved* in C iff either p or $W-p$ is a joint commitment; otherwise, p is *unresolved* in C .
- (66) p is *controversial* in C iff $W-p$ is a commitment of at least one discourse participant, p is unresolved in C , and C is not empty.

A second, and more directly relevant, type of situation in which q is a public belief without being a mutual one is the following. Suppose A has spoken favorably about cats, as before, and let q again stand for the proposition expressed by *Cats make better pets than dogs*. Then q is a commitment of A , according to (63). Consider the state of the discourse before B makes any response indicating agreement or disagreement (i.e., neither q nor $W-q$ is a commitment of B). q is not a mutual belief in this situation, though it may become one without further ado if B indicates acceptance. $W-q$ is not a mutual belief, either, but its status is different from that of q . While q just needs ratification by B to become a mutual commitment, $W-q$ is not eligible as a mutual belief at all, given that A has already expressed commitment to q . In an obvious way the context is *biased toward* q ; only q can be admitted as a mutual belief without requiring (non-monotonic) revision.

This simple and intuitive notion of contextual bias is what I will build upon in accounting for the bias of declaratives. The relevant definitions are given in (67)-(68).

(67) *C* is *biased toward p* iff *W-p* is controversial in *C* and *p* is not controversial in *C*.

(68) *C* is *neutral with respect to p* iff neither *p* nor *W-p* is controversial in *C*.

These definitions are quite straightforward. Contextual bias exists if mutual agreement on *p* is possible (without revision) while mutual agreement on *W-p* is ruled out due to an existing commitment to *p* by at least one discourse participant. If the context is in a neutral state with respect to *p*, then mutual agreement on either *p* or *W-p* is possible in principle.

In effect, the notions of contextual bias and neutrality are realized in terms of accessibility between contexts. A context is classified as biased or neutral depending on what other kinds of contexts can be reached from it, given particular assumptions about how the discourse is expected to progress. This underlying idea can be made more explicit by giving the accessibility relation involved. The accessibility relation encodes the default expectation that the discourse will proceed in a way that is consistent with the commitments already in force for each participant. The expectation of consistency means that a participant can make further moves that *reduce* her commitment set to a (possibly improper) non-empty subset of the current one, a restriction expressible in terms of the power set of the current commitment set. Moves that result in the addition of worlds are unexpected,

hence inaccessible. The definition of the accessibility relation that achieves this effect is given in (69):

- (69) Let R be an accessibility relation between contexts C, C' such that $\langle C, C' \rangle \in R$ iff $cs_A(C') \in \wp(cs_A(C))$ and $cs_B(C') \in \wp(cs_B(C))$ and C' is not empty.

Given a context C , the accessibility relation R defines a set of contexts (reductively) accessible from C , which I call the *reduction set* of C and refer to using the function notation $\mathfrak{R}(C)$:

- (70) $\mathfrak{R}(C) = \{ \langle C, C' \rangle \text{ such that } \langle C, C' \rangle \in R \}$

Note that the trivial reduction, where no worlds are eliminated from either commitment set, is included in the reduction set. Empty contexts are excluded.

The definitions of controversiality, bias, and neutrality given above can be equivalently expressed in terms of the reduction set, as exemplified for contextual bias in (71):

- (71) C is *biased toward* p iff both of (a) and (b) hold:
a. $\exists C' \in \mathfrak{R}(C)$ such that $cs_A(C') \cup cs_B(C') \subseteq p$.
b. $\neg \exists C' \in \mathfrak{R}(C)$ such that $cs_A(C') \cup cs_B(C') \subseteq W-p$.

However, as the simpler definitions in (63)-(68) are perfectly adequate for present purposes, I will not go through the process of replacing them at this time.

For now, the reduction set is useful in two ways. First, it gives us a convenient way to represent the set of possible contexts $\mathbb{C}$ for A and B :

- (72) $\mathbb{C}_{\{A,B\}} = \mathfrak{R}(\langle W, W \rangle)$

(72) is relativized to A and B. The set of all possible contexts, across individuals, would include the contexts for all pairs of individuals eligible as discourse participants.

Second, and more importantly, making the accessibility relation explicit serves to highlight a departure I will make from standard assumptions about the basic goals of discourse. In modeling discourse in terms of *mutual* commitments, it is very often assumed for simplicity (and sometimes rationalized by appeal to Gricean principles) that discourse participants have the common goal of increasing their *mutual* knowledge base. In the possible-worlds framework, an increase in mutual knowledge corresponds to a reduction of the set of worlds of which the mutual beliefs of the participants hold true. The overall goal can thus be characterized as reducing that set of worlds, which is here expressed as $cs_A(C) \cup cs_B(C)$ (the context set, in Stalnaker's terms). I do not rely on such an assumption. All I assume is that each discourse participant, when he or she commits to a position, does not anticipate retreating from or revising that position as the discourse proceeds. That is what *commitment* amounts to. Committing to one's own position does not preclude the goal of getting another participant to revise *their* position to agree with one's own. But it will be mutually understood that the path of least resistance for any individual is maintaining consistency with their own commitments. This is exactly what the accessibility relation expresses: the contexts

reductively accessible from the current one are those in which both participants maintain such consistency.

In closing this section, let me point out that the contextual states introduced here are not particular to the analysis of rising declaratives. They offer a general way to talk about bias and neutrality that is potentially useful for the analysis of other phenomena as well, such as tag questions, negative polar interrogatives, and discourse particles. Furthermore, the types of discourse situations characterized by these states are basic, and ought to be comprehended by a theory of discourse. The idea of contextual bias is simple enough to be implementable in a variety of frameworks besides the one chosen here. The conceptual architecture of the proposal can thus be distinguished from the details of the contextual representation chosen. The next step in the present analysis, however, is to proceed with the details in this framework, specifying the contribution of rising and falling declaratives in terms of their context change potential and linking their effects to the contextual states defined above.

3.2 Declarative meaning and locution meaning

The basic idea to be implemented in this section, following the tradition of update semantics, is that the meaning of a sentence is its context change potential, its CCP (Heim 1982 and others). The modification made here is that the CCP of a sentence is defined in terms of an update to a substructure of the context, the

commitment set (cs) of an individual participant. Consistent with the guiding hypotheses in (54) and (55), the rise and fall will serve to identify the individual cs to be updated, given an utterance context, i.e., a context in which individual participants can be identified in the roles of Speaker and Addressee.

I will use the term *locution*, abbreviated L or $\hat{\updownarrow}S$, to designate the linguistic expression comprised of a sentence of a given type plus the rise or fall, retaining the more traditional usage of the term *sentence* for expressions not specified for intonational category. This and other notational conventions are summarized in (73).

(73) Notation

- a. $\uparrow S_{\text{decl}}$: rising declarative locution
- b. $\downarrow S_{\text{decl}}$: falling declarative locution
- c. $\uparrow S_{\text{interr}}$: rising polar interrogative locution
- d. S: ranges over $\{ S_{\text{decl}}, S_{\text{interr}} \}$
- e. L, $\hat{\updownarrow}S$: ranges over $\{ \uparrow S_{\text{decl}}, \downarrow S_{\text{decl}}, \uparrow S_{\text{interr}} \}$
- f. $\hat{\updownarrow}$: ranges over $\{ \uparrow, \downarrow \}$
- g. cs_X : ranges over $\{ cs_A, cs_B \}$

$\uparrow S_{\text{decl}}$, $\downarrow S_{\text{decl}}$, and $\uparrow S_{\text{interr}}$ may be considered logical representations of locutions with the given syntactic types and intonational categories. I comment further on this point at the end of the section.

The CCP of a declarative sentence is defined with respect to an individual cs_X , as in (74), without regard to the identity of X. The descriptive content corresponds to the proposition expressed by the declarative.

CCP of declarative

(74) $cs_X + S_{\text{decl}} = \{ w \in cs_X : \text{the descriptive content of } S_{\text{decl}} \text{ is true of } w \}$

The CCPs associated with rising and falling locutions are represented in (75) and (76), respectively:

- (75) $C + \uparrow S = C'$ such that:
- a. $cs_{Addr}(C') = cs_{Addr}(C) + S$
 - b. $cs_{Spkr}(C') = cs_{Spkr}(C)$
- (76) $C + \downarrow S = C'$ such that:
- a. $cs_{Spkr}(C') = cs_{Spkr}(C) + S$
 - b. $cs_{Addr}(C') = cs_{Addr}(C)$

Here, cs_{Addr} is a function from an utterance context C to either cs_A or cs_B , depending on who is in the role of Addressee when the locution is uttered.

Similarly, cs_{Spkr} is the function picking out either cs_A or cs_B based on who is in the role of Speaker.⁵ The contribution of the intonation is to identify the target commitment set for the substantive update.

Combining the elements above, we arrive at the result in (77) for a rising declarative locution, applying the declarative update to the Addressee's cs :

CCP of rising declarative

- (77) $C + \uparrow S_{decl} = C'$ such that:
- a. $cs_{Spkr}(C') = cs_{Spkr}(C)$
 - b. $cs_{Addr}(C') = cs_{Addr}(C) + S_{decl}$

The counterpart to (77) for falling declaratives is:

CCP of falling declarative

- (78) $C + \downarrow S_{decl} = C'$ such that:
- a. $cs_{Spkr}(C') = cs_{Spkr}(C) + S_{decl}$
 - b. $cs_{Addr}(C') = cs_{Addr}(C)$

⁵ It should be clear that the meaning I posit for the rise and fall is indexical in nature, on a par with the meaning of expressions such as *I*, *you*, *the Speaker*, and *the Addressee* and amenable to the same sort of analytical treatment.

The only difference between the two CCPs is the target of the substantive update – the Addressee cs in (77), the Speaker cs in (78). In each case the declarative component makes the same contribution: eliminating worlds from the target cs of which the descriptive content is not true.

The compositional nature of the above approach emerges more clearly when the individual functions corresponding to each element are sorted out. A sentence meaning is an update function on commitment sets (sets of worlds), as summarized in (79). As stated in (80), a rise or fall is interpreted as a function taking a sentence meaning and mapping it to a context update function that applies the sentence meaning function to an individual commitment set of the context, leaving the other commitment sets unchanged. Locution meaning follows compositionally, as shown in (81), by applying the intonational function to the sentence meaning.

(79) $|S|$ = function from cs_X to cs_X'

(80) $|\hat{\diamond}|$ = function from sentence meanings (functions from cs_X to cs_X') to functions from C to C' such that cs_X of $C' = |S|(cs_X)$ and C' is otherwise identical to C .

(81) $|L| = |\hat{\diamond}|(|S|)$ = function from C to C' such that cs_X of $C' = |S|(cs_X)$ and C' is otherwise identical to C .

I use the $+$ notation for both kinds of update (cs and C), so that the following equivalences hold:

(82) a. $C + L = |L|(C)$
b. $cs_X + S = |S|(cs_X)$

I make the following provisional assumptions about presuppositions in this framework. Following Heim 1983, I assume that updates are partial functions, defined only for contexts in which presuppositions are satisfied. A context is said to

admit a sentence only if the presuppositions of the sentence are met in that context, which means, in effect, that the presuppositions must be already entailed (or in practice, accommodatable). I extend the notion of admittance to apply to locutions and contexts as well as to sentences and commitment sets, as defined in (83)-(84). I assume that the locution inherits the presuppositions of the sentence. Finally, I assume that presuppositions must be satisfied with respect to *joint* commitments rather than individual sets, as (84) provides.

(83) cs_X admits S iff for all propositions r such that r is a presupposition of S , $cs_X \subseteq r$.

(84) C admits $\hat{\Downarrow}S$ iff for all cs_X in C , cs_X admits S .

The operations $cs_X + S$ and $C + L$ are defined only if cs_X admits S and C admits L , respectively. Rising and falling intonation are assumed to carry no additional presuppositions themselves (although there is no reason why they could not). The treatment of presupposition suggested here distinguishes between presuppositional content and primary descriptive content in a way that seems correct: presuppositions cannot be controversial with respect to the context, as the propositional content proper of a declarative can be.⁶

Consistency is defined at both the sentence and the locution level:

Consistency

(85) S is consistent with a set of worlds cs iff cs admits S and $cs + S \neq \emptyset$.

(86) L is consistent with C iff C admits L and $C + L$ is not empty.

⁶ I speculate that this distinction may also be useful in characterizing the type of content identified by Horn (2000, 2001) as ‘assertorically inert’, which, like presuppositions, may not be controversial. The difference between presuppositions and assertorically inert material could be represented in this framework as the difference between content that must already have the status of a joint commitment (a presupposition) at the time of utterance vs. content that may be novel as long as it’s not controversial in the context, in the sense of *controversial* defined in (66).

Sentences and locutions can be consistent with commitment sets and contexts, respectively, only if their presuppositions are met.

I have simplified above, and will continue to do so, by making direct reference to syntactic sentence type and intonational category in defining CCPs. There is nothing in the above treatment, however, that is incompatible with the more standard procedure of translating a structural object into a logical representation that then forms the input for interpretation. (By ‘structural object’ I mean an object comprising both lexical-syntactic structure in the usual sense and intonational structure – here, the intonational category.) Under such assumptions, S_{decl} and S_{interr} can be viewed as logical representations of sentences with the designated syntactic types. Similarly, $\uparrow S$ and $\downarrow S$ can be understood as logical forms for locutions of the designated intonational categories. I have chosen to use the labels *decl* and *interr*, together with the iconic symbols $\uparrow$ and $\downarrow$, to highlight my claim that the structural types under consideration in this thesis map transparently to semantic categories, and in effect determine the interpretation.⁷ A syntactically declarative sentence will ultimately receive the interpretation in (74), whatever intermediate steps are carried out. A rising locution will ultimately be interpreted as in (75); and so on for the other categories.

⁷An alternative choice, for example, would be to identify the relevant semantic categories as something like ‘commitment mode’ for declaratives, using a form such as S_{commit} , or, in the style of Groenendijk 1999, $\theta!$ for declaratives and $\theta?$ for interrogatives. Similarly, rising and falling intonation could be represented logically as the *Addr* and *Spkr* operators, respectively.

3.3 Interrogative meaning

To frame the contrast between interrogatives and rising declaratives, I will now give a proposal for the meaning of rising interrogatives in context update terms. Consistent with the general hypotheses about the rise given in (75), I assume that an interrogative sentence with rising intonation operates on the commitment set of the Addressee. Interrogatives, however, do not commit the Addressee to their content, or the Speaker either. In fact, interrogatives seem to differ crucially from declaratives in not expressing commitment at all. The CCP of an interrogative is thus very simple – it makes no change to the targeted commitment set. The definition is given in (87); (88) shows the compositional result for the CCP of a rising interrogative.

$$(87) \quad cs_X + S_{\text{interr}} = cs_X$$

$$(88) \quad C + \uparrow S_{\text{interr}} = C' \text{ such that:}$$

$$\text{a. } cs_{\text{Addr}}(C') = cs_{\text{Addr}}(C) + S_{\text{interr}}$$

$$\text{b. } cs_{\text{Spkr}}(C') = cs_{\text{Spkr}}(C)$$

Effectively, $C + \uparrow S_{\text{interr}} = C$ (ignoring effects of any accommodated presuppositions). A significant advantage to treating interrogative updates in the same manner as declarative ones is that the assumptions about presuppositions made in (83)-(84) apply straightforwardly to interrogatives. Since polar interrogatives carry the same presuppositions as their declarative counterparts, this is a desirable result.

The above definitions do not do full justice to our intuitions about interrogatives. Intuitively, a context in which an interrogative has been uttered is

not identical to one lacking the interrogative utterance. I assume that polar interrogatives do have non-trivial effects on aspects of the discourse context not represented here. For example, Buring 1995 and Roberts 1996, following Carlson 1983, explicate the effect of interrogatives in terms of the Discourse Topic, or Question Under Discussion (QUD), a structure paralleling the CG. Groenendijk 1999 represents the non-trivial update effects of interrogatives with a more complex representation of the context using pairs of worlds. Proposals agree, however, on the point that matters to the present discussion — polar interrogatives do not commit any participant to their descriptive content.

By the *descriptive content of an interrogative* I mean just the proposition expressed, in an obvious way — e.g., *It is raining* for the interrogative *Is it raining*. A negative polar interrogative will thus have different descriptive content than its positive counterpart. Most semantic theories do not give polar interrogatives a representation that distinguishes between *Is it raining* and *Isn't it raining*, as this assumption requires. However, the failure to distinguish between the two is a shortcoming that needs a remedy independently of any proposal made here, given that the effects of positive vs. negative polar interrogatives are manifestly not identical. I will therefore continue to refer to the descriptive content of interrogatives, assuming that the logical representation of an interrogative provides that content.

Since interrogatives by hypothesis do not change the context, the definition of consistency given in (86) ensures that they are trivially consistent with any

context that is non-empty to begin with.⁸ Of particular interest for present purposes is the observation that a rising interrogative with descriptive content *p* is consistent with a (non-empty) context in which the Addressee is already committed to *p*. Intuitively, an interrogative seems to be redundant in such circumstances. Given that the Addressee's commitment to *p* is already a matter of public record, the interrogative seems to be calling for a response that has already been provided. It is common practice in modeling discourse to focus on informativeness in terms of the literal effect of utterances, and to assume that participants adhere to principles that require them to be informative in this sense. Hence, many models of discourse (including those mentioned above) incorporate rules that prohibit redundant statements, along with barring interrogatives whose answers would be redundant. For the present account, however, it is crucial that the redundancy of an utterance be understood as distinct from its felicity. (See Section 3.5 for further relevant discussion.) The fact is that what we may think of as 'reiterative questions', i.e., questions to which an answer has already been given, *are* felicitous. Examples like (45)- (46) from Section 2.3, repeated below, demonstrate that:

- (45) A: That copier is broken.
 B's response:
 a. Is it? Thanks, I'll use a different one.
 b. It is? Thanks, I'll use a different one.
 c. (Oh), it is. Thanks, I'll use a different one.

⁸The consistency condition may be too liberal in allowing interrogatives to be consistent with any non-empty context at all. As Büring and Gunlogson 2000 note, there seem to be restrictions on the use of polar interrogatives in contexts biased *against* *p*. However, such contexts will not concern us here, and motivating revisions for interrogatives would take us too far afield from the analysis of rising declaratives.

- (46) A: Jake's here.
 B's response:
 a. Is Jake here? Then let's get started.
 b. Jake's here? Then let's get started.
 c. (Oh), Jake's here. Then let's get started.

We will see more examples of this sort in the next chapter, where the functioning of declaratives as questions will be linked to the felicity of rising interrogatives in contexts like those illustrated in Section 2.3.

3.4 Locutionary bias and neutrality

Locutionary and sentential updates can be classified in various useful ways according to their effects on commitment sets and contexts. In particular, now that we have a notion of contextual bias (from Section 3.1) and a proposal for declarative meaning (Section 3.2), we are in a good position to characterize the systematic effects of declarative locutions on the contexts in which they occur. The bias or neutrality of a locution can be derived from the corresponding contextual states, as (89)-(90) state.

Given a locution L with descriptive content p and a context C that admits L :

- (89) L is neutral with respect to C iff $C + L$ is neutral with respect to p .
 (90) L is biasing with respect to C iff C is neutral with respect to p and $C + L$ is biased toward p .

It follows from the definition of the CCP of declaratives, together with the definitions in (89)-(90), that no use of a declarative can be neutral. There are two types of outcome when a declarative is uttered in a context with which it is

consistent. Either the declarative will eliminate worlds of which its content is not true from the cs of some participant, resulting in a state of controversy or contextual bias; or the declarative will have no effect. A declarative can only be without effect, however, if the context is already non-neutral, that is, if worlds of which the content is not true are already absent from the targeted cs.

Similarly, it follows from the definition of interrogatives as *not* changing the commitment set that no occurrence of an interrogative can be biasing. An interrogative is simply not suitable for the job of introducing bias into a neutral context. The generalizations in (91)-(92) thus have the status of theorems:

Theorems

(91) No $\Updownarrow S_{\text{decl}}$ is neutral with respect to any C.

(92) No $\Updownarrow S_{\text{interr}}$ is biasing with respect to any C.

Note that it does not follow from the proposal that all declaratives are biasing, given the above definitions, nor that an interrogative locution is always neutral. A locution is biasing only if it effects a transition from a neutral to a biased context. In contexts where a declarative locution is uninformative, it cannot be biasing.

Similarly, interrogatives qualify as neutral only when uttered in a neutral context. An interrogative used in a biased context preserves the bias, and thus does not count as neutral. But an interrogative, unlike a declarative, has the capacity to preserve neutrality as well. It is thus possible for an interrogative to be neutral, whereas a declarative does not have that potential.

The idea that declaratives express commitment to their content can now be given shape as well.

Commitment as a property of locutions (across contexts)

- (93) L *expresses commitment* to p iff there is no C such that L is consistent with C and C + L is neutral with respect to p.

Expressing commitment is a characteristic property of declarative locutions, both rising and falling varieties. Interrogatives do not express commitment.

The unacceptability of declaratives in the contexts illustrated in Section 2.2 follows directly from (91). By hypothesis, those examples involved discourse situations requiring the Speaker to maintain (at least the appearance of) a neutral context with respect to the issue raised by the question. Use of a declarative to express the question, whatever the intended function of the move, is guaranteed to result in a non-neutral context, in violation of the expectation of neutrality. On the other hand, the capacity for interrogatives to be neutral accounts for their acceptability in the same set of examples.

The noncommittal nature of interrogatives is also supported by the examples in Section 2.3, which showed that interrogatives as well as rising declaratives failed to commit the Speaker to their content. The differences between rising and falling declaratives in Section 2.3 also follow; a representative example is repeated below:

- (44) [A&B are looking at a co-worker's much-dented car]
A: His driving has gotten a lot better.
B's response:
a. Has it? I don't see much evidence of that.
b. It has? I don't see much evidence of that.
c. It has. #I don't see much evidence of that.

Since the Speaker uttering a rising declarative does not commit herself to the content uttered, the range of attitudes shown to be possible throughout Section 2.3

is perfectly consistent with the proposal. In particular, the Speaker may, without inconsistency, go on to follow the rising declarative with an expression suggesting skepticism or disagreement, as (44) shows. Since the speaker uttering a falling declarative does commit herself, the contrast between the two is understandable. The speaker uttering (44c) is behaving inconsistently, whereas the speaker of (44b) is not.

3.5 *Entailment, uninformativeness, and vacuousness*

In this section I provide definitions that will be useful in understanding and classifying the effects of various locutions in the contexts in which they are uttered. A basic notion is that of entailment. Entailment is defined as a relation between a context and a locution, as in (94), or between a sentence and a commitment set, as in (95).

Entailment

(94) $C \models L$ iff C admits L and $C + L = C$

(95) $cs \models S$ iff cs admits S and $cs + S = cs$

The idea behind the definitions in (94)- (95) is simply that a context entails a locution when the effect of that locution has already been achieved. Similarly, a commitment set entails a sentence when updating the commitment set with that sentence has no effect.

Thus, a declarative sentence will be entailed by a commitment set only if the content of the declarative is already a commitment in that set. A rising or falling

declarative locution is entailed by a context C only if the commitment expressed by the declarative is already a commitment of the Speaker or Addressee (depending on the intonational category). If the declarative registers a new commitment for some participant, it is not entailed by the context.

A consequence of (95), together with the CCP of interrogatives given in (87)-(88), is that an interrogative sentence is entailed by any commitment set that admits it, and a rising interrogative locution is likewise entailed by any context that admits it. Again, this is clearly not the right result for interrogatives in the long run. We would like to allow for a distinction, e.g., between an interrogative uttered for the first time in a context and the repetition of the same interrogative, just as there is a distinction between an initial utterance of a declarative locution (which isn't necessarily entailed by the context) and its subsequent repetition (when the locution is entailed). The current representation of the context does not allow such distinctions between interrogatives to be captured. But for the present proposal, which involves the contrast between effects of declaratives vs. interrogatives, this shortcoming will do no real harm.

A notion that is going to play an important role in the next chapter is that of *uninformativeness*. In (96)-(97) informativeness and uninformativeness are defined as relations between locutions and individual commitment sets.

(Un)informativeness

(96) $\hat{\Downarrow}S$ is informative with respect to $cs_X(C)$ iff C admits $\hat{\Downarrow}S$ and $cs_X \not\models S$.

(97) $\hat{\Downarrow}S$ is uninformative with respect to $cs_X(C)$ iff C admits $\hat{\Downarrow}S$ and $cs_X \models S$.

Derivatively, a locution may be defined as informative or not with respect to the context as a whole:

- (98) $\Updownarrow$ S is informative with respect to C iff it is informative with respect to at least one cs in C.
- (99) $\Updownarrow$ S is uninformative with respect to C iff it is uninformative with respect to every cs in C.

Under the definitions in (96)-(97), a locution that is entailed by a context may still be informative with respect to that context. That is because (un)informativeness is defined in terms of the *potential* effect of S on a commitment set, without regard to whether the locutionary update actually operates on that cs or not. A speaker may, for example, repeat a falling declarative statement made previously in an attempt to convince a skeptical interlocutor to accept its content. The locution is entailed by the context when repeated but still informative with respect to the Addressee's commitment set, and therefore, by (98), with respect to C.

To make this point clear, consider the following exchange:

- (100) A [watching a bird fly away]: That was a kingfisher.
B: Are you sure? It looked like a seagull to me.
A: I'm positive. It was a kingfisher.

Assuming that neither A nor B have prior commitments as to the identity of the bird in question,, A's first statement, *That was a kingfisher*, is informative with respect to both the Speaker's and the Addressee's commitment sets, and also informative with respect to the context as a whole. The update represented by the declarative sentence would result in a change if applied to either set. But as a *falling* declarative, the locution can only effect an *actual* change to the Speaker's

commitment set; by definition a falling declarative leaves the Addressee's commitment set untouched. The falling declarative does effect a change in the context of its utterance, because it adds a new commitment to the Speaker's set. Thus, by (94), the locution is not entailed by the context in which is uttered.

A's second turn includes the statement *It was a kingfisher.*, which serves to reiterate an existing commitment of A's. This time around, *It was a kingfisher.* is uninformative with respect to the Speaker's commitment set, since A has already committed to the content, but it is still informative with respect to the Addressee's, since B hasn't committed to that same content. In this second case the falling declarative locution is entailed by its context of utterance. It has no effect on the Speaker's commitment set, since the Speaker is already committed. And it has no effect on the Addressee's commitment set, by definition. But by (98), *It was a kingfisher* is still informative with respect to the context. It satisfies (98) by being informative with respect to the Addressee's commitment set. Thus, a locution used to insist on one's position, as A does, qualifies as informative as long as that position has not yet become a joint commitment.

Interrogatives, according to the above definitions, turn out to be uninformative with respect to any context that admits them. That is because the CCP of an interrogative is defined as an identity function. An interrogative thus has no potential for updating a commitment set, whoever that set belongs to. Unlike the results for entailment, the classification of interrogatives as inherently uninformative is the right result. It matches the intuition that interrogatives do not

directly affect commitments, either the Speaker's or the Addressee's. Lacking the potential to alter commitments, interrogatives are not a natural vehicle for stating and/or reiterating one's position.

Uninformativeness with respect to the context is thus the norm for an interrogative locution. But a declarative locution that is uninformative with respect to the context as a whole (as opposed to just an individual commitment set) is a strange sort of beast. For a declarative locution to qualify as uninformative with respect to the context, it must be the case that its content is already a commitment of both participants, i.e., a joint commitment. Intuitively, the declarative is pointless in such circumstances. There is nothing to be gained by uttering it. To describe this sort of case I introduce one final notion, that of *vacuousness*:

(101) L is vacuous with respect to C iff p is a joint commitment in C, where p is the descriptive content of L.

(101) classifies a locution as vacuous if its content is already positively resolved in the context. Any locution that is vacuous with respect to a context will also be uninformative with respect to the context and entailed by it. But an uninformative locution is not necessarily vacuous. In particular, interrogatives aren't necessarily vacuous just because they are uninformative. An interrogative, intuitively, raises the issue of whether its descriptive content is true. Although we have no explicit way of modeling this issue-raising aspect of interrogatives at the moment, it is clear that raising an issue may be a very useful step if one's goal is to resolve it, or even simply to discuss it. The one kind of case in which an interrogative locution, or any

other locution, *does* qualify as vacuous, according to (101), is where the content is not even an issue that warrants discussion – where the matter is already settled in the context.

I will employ the notions defined in this section, and in particular the notion of uninformativeness with respect to a commitment set defined in (97), to understand how the questioning use of declaratives arises. It is important to note that these definitions play a fundamentally different kind of role in my account than they do in game-oriented models of discourse (e.g., Groenendijk 1999), which typically incorporate rules prohibiting moves of certain sorts. By contrast, I do not prohibit the use of declarative locutions that are uninformative or entailed in a context. Rather, to the extent that speakers do make such moves, I assume that they do so intentionally. I take it that the interaction between locution meaning and context is part of what speakers understand about their language – part of their semantic competence. Knowing that their interlocutors possess this competence as well, speakers may rely on the interaction of context and locution to provide clues as to their intentions. The definitions I give here are simply a classificatory aid in understanding those clues, providing labels for certain types of interactions between locutions and contexts.

I will not prohibit the use of vacuous locutions either, even though they have a different status than the other categories. Utterances of vacuous locutions are systematically anomalous in a way that uninformative and entailed utterances are not. I assume that a principled explanation for this anomaly can be offered on

general grounds. Vacuous discourse moves are unexpected in the same way that performing any sort of task whose potential effect has already been achieved is an unexpected move for a rational agent.

To illustrate these points, consider the contrast between (102) and (103). In (102), A states that there's nothing in the refrigerator; B may respond felicitously with any of the three locution types. In (103), it is B who remarks on the emptiness of the refrigerator, and A agrees. Here B's response, in whatever form, is vacuous.

(102) A: I've just searched the refrigerator and there's absolutely nothing cold to drink.

B:

- a. Are we out of beer?
- b. We're out of beer?
- c. (So) we're out of beer.

(103) B: I've just searched the refrigerator and there's absolutely nothing cold to drink.

A: Yeah, I know. We're out of just about everything.

B:

- a. #Are we out of beer?
- b. #We're out of beer?
- c. #(So) we're out of beer.

In (102), B's question is not vacuous, even if understood as asking specifically about beer in the refrigerator. (Another possibility is that B is understood as asking about a supply of beer stored elsewhere, say on the back porch.) While A's statement seems to commit A to the entailment that there is no beer in the refrigerator, B's question is still nonvacuous since the entailment does not, at the time of B's utterance, have the status of a *joint* commitment. Note that the rising

declarative and interrogative qualify as both uninformative and entailed in the context of utterance, but are perfectly felicitous.

In (103), by contrast, B's question, whatever form it takes, is peculiar because vacuous, given that A and B agree about the barren state of the refrigerator. The question is odd because it raises an issue that is apparently already resolved in the context. In fact there is a strong tendency to interpret the question in (103) so as to avoid the anomaly associated with vacuousness by giving B's question the alternate interpretation mentioned above, taking it to ask about beer that is stored somewhere other than the refrigerator. Under that interpretation (103a) is fully acceptable. The declarative versions lend themselves to this interpretation, however, only if A's statement is also understood as applying to supplies generally, not just those stored in the refrigerator. The explanation of that restriction will have to await the developments of the next chapter.

3.6 Operating on commitment sets

In this dissertation, which focuses on questions, the property of uninformativeness defined in the previous section is of more concern than informativeness. To preview the argument of the next chapter, the claim I will defend with respect to declaratives is that they are interpretable as questions only when uninformative with respect to the Addressee's commitment set. If this claim is correct, declaratives should be interpretable as questions only when the

Addressee is already publicly committed to the content of the declarative. The data to be introduced in Section 4.2 will show that this generalization is accurate.

Restricting attention to uninformative rising declaratives allows me to sidestep the issue of what it means for a speaker to make a move that has the effect of modifying the Addressee's public commitments, as would be the case for an informative rising declarative. I do assume that such moves are possible, and that a coherent account can be given of them. This is the door I will leave open for pursuing an account of the informative uses of rising declaratives exemplified earlier in (31b) and (32b) (see p. 27). I will not, however, embark on that account in this dissertation.

Still, a few comments are in order about the notion of operating on commitment sets generally, and on the Addressee's set in particular. There is one way in which this is already a routine and widespread notion. It is common practice to formalize assertion as an operation adding the asserted proposition to the Common Ground, or equivalently, reducing the context set. (See, e.g., Heim 1982, Krifka 1991; note that Stalnaker 1978 does not take this step.) The idea of operating on the Addressee's public commitments is embedded in this characterization of assertion, since shared assumptions can only be updated via the update of the Addressee's as well as the Speaker's public assumptions. Following Stalnaker, we may assume that the asserted content becomes part of the Common Ground only if there are no objections made by the Addressee. In similar fashion,

the success of an operation on the Addressee's commitment via a rising declarative may be assumed to be contingent on the Addressee's acceptance of it.

In the long run, however, this is not a completely satisfying response. The concern about operating on commitment sets is not allayed so easily, particularly since the present proposal associates such operations not with speech act categories like assertion, where acceptance and rejection of a particular act have some plausibility, but with sentence and locution meaning, which is at a much more fundamental level. As I have already explained, for the purposes of dealing with questioning uses of declaratives these concerns are not pressing (at least, no more pressing than usual). In the next chapter I will proceed with the rest of the analysis without worrying about them. But in the remainder of this section I offer a few speculative remarks about the nature of operations on commitment sets.

One point to be kept in mind is that a commitment set is not a direct representation of the doxastic state of an individual. It is rather a representation of the individual's *public* commitments, as was stressed at the beginning of this chapter. Let us consider for a moment what it means for a Speaker to register a change to his own commitment set. Presumably the act of publicizing a commitment does not coincide with the actual formation of that belief or assumption but bears some more indirect relationship to it. It can, perhaps, be viewed as a report of the Speaker's state, or a description of the Speaker's (formerly private) belief or assumption that is made public by the speech act.

If we try to conceptualize in the same *descriptive* terms an operation on the Addressee's commitment set performed by the Speaker, it seems very strange indeed. The Speaker as a rule is in no position to publicize the Addressee's private beliefs or assumptions. And even if he were, such a description of the Addressee's state would not be a good way to induce a change in or addition to the Addressee's beliefs; it would just be a description of one of them. It would be something like saying to the Addressee, "You believe such-and-such." There may be a point to such a move, and there may in addition be nontrivial side effects that the Speaker intentionally exploits. But it does not on the whole strike me as a promising starting point for understanding informative rising declaratives.

If I were going to tackle the use of informative rising declaratives in this thesis, I would start by rejecting the descriptive view just outlined, for both kinds of commitment sets. The alternative to be pursued is understanding an operation on commitment sets as a kind of modal, or dispositional, meaning, in a more or less literal sense – meaning that has to do with the disposition (in the sense of *placement*) of the sentence in the discourse structure. Such meaning is not descriptive in any ordinary sense. The connection between a commitment set update and the epistemic state of the designated participant is not direct, though we can expect, given Gricean principles, a certain amount of correspondence between what the Speaker chooses to do to his own commitment set and his own internal state. As for operations on the Addressee's commitment set, the implications would need to be sorted out carefully in the context of a particular proposal. But at least it

would be possible in principle for a Speaker to make a move that *anticipates* (or attempts to achieve) commitment by the Addressee rather than describing it.

For this kind of approach to work, a principled distinction must be made between the purely epistemic component of the Common Ground and the discourse commitments, which may or may not have an epistemic foundation. It only makes sense to think of operating on *discourse* commitments in the manner just described. Even in the relatively straightforward case of a Speaker committing himself to some content (i.e., operating on his own commitment set), this distinction is useful. However, I will not undertake it here.

For now, I sum up by reiterating that the use of rising declaratives in situations where the content is novel with respect to the Addressee's commitments is not in the domain of this thesis. I claim in the next chapter that such uses cannot constitute questions, and questions are what I am primarily concerned with. However, I do maintain that a virtue of the present analysis is that it suggests a way in which the informative use of rising declaratives might eventually be reconciled with their questioning use.

Chapter 4

Questioning

4.1 Uninformativeness and questioning

So far we have reached an understanding of the non-neutrality of declaratives that accounts for important aspects of their interpretation. The challenge that remains is accounting for the questioning use of rising declaratives. Under the proposal advanced in the previous section, rising declaratives update the Addressee's commitment set. Although the bias of rising declaratives becomes comprehensible on this view, it is not obvious on the face of it why a move by the Speaker to update to the Addressee's commitments should naturally be interpreted as questioning. Nor, it turns out, is the characterization of declarative bias sufficient to account for all the restrictions on their use as questions. In this chapter I'll document the additional restrictions on use and connect them to the questioning use of rising declaratives. I show how the questioning use of rising declaratives can be derived, together with the restrictions, from the account of their meaning already given, together with certain assumptions about questioning.

The position I defend is that declaratives, even rising ones, are not inherently questions, a position compatible with the account so far. Rather, as will be shown in Section 4.2, they operate as questions only in certain restricted contexts. The

contextual requirement is that the proposition expressed by the declarative must already be known to the Addressee, and mutually known to be known – i.e., it must be a public commitment of the Addressee. Intuitively, for declarative questions, the idea is that the context must provide some crucial information about how the Speaker expects his use of the declarative to be interpreted. Declaratives can be interpreted as questioning moves only if they *can't* be interpreted as telling. The phenomenon of rising declarative questions thus arises as an interaction between locution meaning and context. Interrogatives, on the other hand, *are* inherently questions and need no special contextual support to acquire this function. (Recall that the term *interrogative* is used exclusively for a syntactic sentence type, while *question* and *polar question* name a category of utterances.)

The hypothesis I will use to guide the analysis is that uninformativeness with respect to the Addressee is a necessary condition for an utterance of a given locution to qualify as a polar question. An utterance of an interrogative will meet the condition in any context that admits it, since interrogatives are uninformative by definition. In fact, it is really the category of interrogatives that supplies the hypothesis, a point to be revisited in Section 4.3, where I will also take up the issue of sufficiency conditions for questioning. I take it that there is a very general sense in which we consider the use of an interrogative, whatever the intent behind the use, to constitute a question. The examples of interrogatives throughout Section 2.2, for example, all are questions in this general sense, although many of them are not (sincere) requests for information.

Let us say, then, that to function as a polar question in the broad sense, an utterance must minimally have the relevant property of a (rising) interrogative: it must be uninformative with respect to the Addressee in the context in which it is uttered. This amounts to the condition on polar questions expressed in (104):

Uninformativeness Condition

(104) An utterance of a locution L is a polar question in C only if L is uninformative with respect to $cs_{Addr}(C)$.

The definition of uninformativeness given in Section 3.5 is repeated for reference:

(97) $\hat{\uparrow}S$ is uninformative with respect to $cs_X(C)$ iff C admits $\hat{\uparrow}S$ and $cs_X \models S$.

Since interrogatives are incapable by definition of altering a commitment set, they are always uninformative with respect to any commitment set. Thus a felicitous utterance of a rising interrogative is guaranteed to satisfy (104). Declaratives are a different matter. Since a declarative is *potentially* informative, whether a particular declarative utterance is informative with respect to the Addressee's commitments or not crucially depends on the context in which it is uttered. In particular, declaratives are uninformative in the required way only in contexts in which the Addressee's public commitments already entail the proposition expressed. In such contexts, since the Addressee is mutually assumed to be committed to p already, the Speaker using the declarative cannot be construed (by the Addressee) as intending to *tell* the Addressee that p holds. My claim is that this public clue to the Speaker's intentions is prerequisite for the success of a declarative question.

In the next section I will support the claim empirically by showing that declaratives can function as questions *only* where they meet the criterion of unformativeness with respect to the Addressee’s commitment set. For clarity of exposition I give a version of the Unformativeness Condition tailored to declarative questions in (105) and call it the *Contextual Bias Condition*:

Contextual Bias Condition on declarative questions

(105) An utterance of $\updownarrow S_{\text{decl}}$ with descriptive content p is interpretable as a polar question in C only if $cs_{\text{Addr}}(C) \subseteq p$.

Keep in mind, though, that the Contextual Bias Condition is not distinct from the Unformativeness Condition in (104) but is just a specialized version of it; (105) is explicit about the circumstances under which a declarative locution satisfies (104).

The Contextual Bias Condition is quite strong in a particular way: it demands that the Addressee’s commitment be a *public* belief. To see this clearly, the Contextual Bias Condition as it stands should be compared to the weaker alternative stated in (106):

Alternative version

(106) An utterance of $\updownarrow S_{\text{decl}}$ with descriptive content p is interpretable as a polar question in C only if it is plausible from the Speaker’s point of view that the Addressee believes p .

The important difference between (105) and (106) is that the latter allows for the Speaker’s *private* information about the Addressee’s *private* belief state to license the rising declarative question, while the former insists that the Addressee’s belief (and hence the Speaker’s awareness of it) must be public.

For the explanation advanced above, which motivates the Contextual Bias Condition in terms of unformativeness, the stronger version is crucial. The reason is simply that the Speaker's private beliefs, being private, cannot give the Addressee the kind of clue to the Speaker's intention that public unformativeness provides. When the Addressee's prior commitment to *p* is mutually recognized, the Addressee can be sure that the Speaker's use of an uninformative declarative is deliberate, and pragmatic reasoning about intentions can proceed accordingly. But if the Speaker's awareness of the Addressee's belief remains private to the Speaker, then the Addressee who is presented with a declarative whose content she already believes cannot be sure (in principle) that the Speaker intended to be uninformative. (In practice, though, there may be other pragmatic clues available as to the Speaker's intent, which will play a role in whether the declarative can be accommodated as a question or not; see the discussion of (119)-(123) below.)

We need not rely exclusively on this conceptual argument in favor of (105), however. As we will see, (105) is independently motivated by the observations to be introduced in the next section.

An important feature of the approach to declarative questions in terms of unformativeness is that the contextual restrictions identified are peculiar to the use of declaratives *as questions*. We do not expect to find declaratives intended as *statements* to be subject to them. The point is abundantly clear for falling declaratives, whose uses are obviously not confined to questioning. The importance for rising declaratives is that it makes less mysterious the existence of examples

like (31)-(32) (seen in Section 2.2, p. 22), where the rising declarative is used to tell the Addressee a piece of news. The analysis cannot be said to predict the existence of those uses, but at least it doesn't predict their nonexistence. In this respect it has a head start on any account in which the rise is directly associated with a questioning function or an attitude of uncertainty on the part of the Speaker.

4.2 *The Contextual Bias Condition on declarative questions*

The data and the empirical generalization advanced in this section center around the observation that declaratives, rising or falling, cannot readily be used as questions 'out of the blue', with no particular context, as interrogatives can be. The first indication is that interrogatives may be used to initiate discourses in ways that the corresponding declaratives may not be. (107a) can be used to strike up a conversation with a stranger about his dog, while (107b-c) are awkward to impossible in the same setting. An interrogative like (108a) may be uttered in the mere hope of an affirmative response, but (108a) seems to presume the hope is justified, which is odd under the circumstances. Similarly, (109a), but not (109b) or (109c), is a standard way of beginning a telephone conversation in the hopes of speaking to Laura.

- (107) [to passerby walking a dog]
Pardon me, but...
a. Is that a Weimaraner?
b. #That's a Weimaraner?
c. #That's a Weimaraner.

- (108) [to passerby in a parking lot]
Excuse me, but my battery's dead.
a. Do you (by chance) have jumper cables?
b. #You've (by chance) got jumper cables?
c. #You've (by chance) got jumper cables.

- (109) [initiating a phone conversation]
a. Is Laura there?
b. #Laura's there?
c. #Laura's there.

It is important to see that the awkwardness of the above declaratives does not follow from the account of declarative bias given in Section 3.4. We expect, in light of that account, that a question asked via a declarative will be non-neutral. But here, unlike the cases exemplified in Section 2.2, there does not seem to be any plausible expectation of neutrality from the Speaker that is violated by the declarative question. Suppose, for instance, that the Speaker of (107b) privately knows or suspects that the dog in question is a Weimaraner. In fact, it is difficult to avoid making that assumption, even for the interrogative in (107a), given that the Speaker has chosen to ask a polar question about a particular breed rather than simply asking what kind of dog it is. Why should it be so odd for the Speaker to ask a non-neutral declarative question in that case, in effect conveying her own positive bias by way of conveying an expectation that the Addressee will agree? Similarly, assume that the Speaker in (109b) has good reason to believe that Laura is indeed home (perhaps he is parked outside Laura's house and calls from his cell phone after he has seen her arrive). Adding this assumption makes it reasonable from the Speaker's point of view to ask a non-neutral question – the Speaker has reason to

expect a *yes* from the Addressee, or at least to object to a *no* – but the assumption does not suffice to make the declarative acceptable.

Further illustration of the point that declarative questions have particular contextual requirements comes from looking at titles (of books, papers, songs, etc.) In (110a)- (112a), we see that interrogatives make reasonable titles; they can introduce a topic without requiring any particular context or assumptions about the targeted reader. Indeed, assuming that a title is designed to be intelligible (within limits) to anyone who happens upon it, this sort of example provides an excellent illustration that (positive) interrogatives require no particular context (beyond the usual one of shared language and culture). Falling declarative titles, as shown in (110c)- (112c), have a similar status in that they are felicitous as statements (though not as questions) without requiring any particular assumptions about the context. The rising declarative titles in (110b)- (112b), by contrast, are decidedly odd. The effect they are intended to produce on the uninitiated reader is not at all clear. To the extent that they are interpretable, they suggest a response to a claim or issue raised earlier, as if the title is intended to be understood as potentially controversial in the context of a discussion already taking place. (Linguists occasionally use such titles with exactly that effect.)

- (110) a. *Must We Mean What We Say?* [philosophy book by S. Cavell]
 b. *#We Must Mean What We Say?*
 c. *We Must Mean What We Say.* [# as a question]

- (111) a. *Can This Marriage Be Saved?* [title of a magazine column]
 b. *#This Marriage Can Be Saved?*
 c. *This Marriage Can Be Saved.* [# as a question]
- (112) a. *Are Rising Declaratives Questions?*[potential title for this dissertation]
 b. *#Rising Declaratives are Questions?*
 c. *Rising Declaratives are Questions.* [# as a question]

Again, there does not seem to be any *a priori* reason why questions used as titles should require a neutral stance. Why shouldn't a rising declarative title be usable to convey both the topic to be discussed and the expectation that the discussion will ultimately be resolved in favor of the proposition expressed? Thus the non-neutrality of declaratives is not by itself a sufficient explanation for the infelicity of rising declaratives in (110)-(112), nor does it account for the strong sense that some prior context is required. The informedness of the Speaker (i.e., the author, in these cases) cannot be the relevant issue either. Nothing bars the assumption that the Speaker has good reason to know or suspect that the proposition expressed is true, but making that assumption doesn't improve the examples.

The examples of felicitous rising declarative questions we saw in Section 2.3 also provide evidence against the idea that the contextual requirement involves the *Speaker's* propositional attitude toward the content of the declarative. Recall that Section 2.3 established the compatibility of rising declarative questions with Speaker attitudes ranging from disbelief to neutrality to positive acceptance, concluding that rising declaratives fail to commit the Speaker to their propositional content. Any attempt to frame the contextual condition on rising declarative

questions in terms that require a particular attitude on the part of the Speaker will immediately fall afoul of those data.

A solution that is compatible with the examples in Section 2.3, however, is that the requirement involves the Addressee's attitude. In all of the cases of felicitous rising declarative questions we have seen so far, the Addressee may reasonably be taken to be committed to the proposition expressed, even when the Speaker is not. Examples (44) and (47) are repeated below to illustrate.

(44) [A&B are looking at a co-worker's much-dented car.

A: His driving has gotten a lot better.

B's response:

d. Has it? I don't see much evidence of that.

e. It has? I don't see much evidence of that.

f. It has. #I don't see much evidence of that.

(47) A: The king of France is bald.

B's response:

a. Is France a monarchy?

b. France is a monarchy?

c. #France is a monarchy.

Intuitively, it seems as though the Speaker must have some evidence to support the belief that the Addressee is committed to the proposition expressed – which evidence comes in the above examples from the Addressee's own statements to that effect. But this view of the situation, though not exactly wrong, is somewhat deceptive. As we will see later in this section, a preceding utterance by the Addressee is not strictly necessary. On the other hand, not just any sort of evidence for the Addressee's belief will suffice. In particular, it is not sufficient for the Speaker to have a private reason for believing that the Addressee believes p. To see

this, return to example (107) and suppose, as was suggested earlier, that the Speaker has good reason to think, based on her own personal experience and knowledge, that the dog is a Weimaraner. She also has good reason to suspect that the owner knows what kind of dog it is; owners of purebred dogs usually do. It seems to follow that the Speaker can reasonably assume that the Addressee believes the dog to be a Weimaraner (though nothing in the context allows the Addressee to reconstruct that reasoning.)

The problem is that adding these background suppositions about the Speaker's private assumptions fails to improve the rising declarative questions. This is not to say that rising declarative questions can never be accommodated in such situations; I will say more on that point later in this section. But to accommodate a declarative question is more complicated than simply assuming that it's reasonable for the Speaker to infer that the Addressee is committed.

For another relevant illustration, recall example (48) from Section 2.3:

- (48) a. Is shoplifting fun?
b. Shoplifting's fun? [# as an out-of-the-blue question]
c. #Shoplifting's fun. [# as insinuating that the Addressee has shoplifted]

The discussion of (48) (see p. 32) described a scenario in which the Speaker knew that the Addressee had shoplifted, but the Addressee was unaware of the Speaker's knowledge. As a conversational opener between acquaintances, (48a) is acceptable (if unkind) in these circumstances, whereas (48b) is infelicitous – an unexpected result if all that mattered were the Speaker's knowledge about the Addressee.

These considerations support the Contextual Bias Condition introduced in Section 4.1, which makes reference to the Addressee's and not the Speaker's commitments, and crucially, the Addressee's *public* commitments. I repeat the condition below for reference:

- (105) An utterance of $\hat{\uparrow}S_{\text{decl}}$ with descriptive content p is interpretable as a polar question in C only if $CS_{\text{Addr}}(C) \subseteq p$.

One of the agreeable consequences of (105) is that the “echo” uses of rising declaratives fall out naturally. The situation where the Addressee has already stated the content presented by the declarative question is just a special case subsumed under the more general condition given in (105). “Echoes” can range from utterances having every word in common with the original, as in (113), to elliptical echoes such as (44) (shown above) and other examples in Section 2.3, to entailments of the content as in (114), which has no word in common with the original.

- (113) A: There's a leopard in the living room.
 B's response:
 a. ? Is there a leopard in the living room?
 b. There's a leopard in the living room?
 c. There's a leopard in the living room.

- (114) A: Gina went skydiving yesterday.
 B: You're kidding!
 a. ? Did she jump out of an airplane?
 b. She jumped out of an airplane?
 c. #She jumped out of an airplane.

Whether or not all of these repetitions are properly called “echoes” is, fortunately, a point of little importance to the present discussion, since that notion plays no

formal role in the account. What's important is that the definition in (105) is general enough to cover all of the cases that might reasonably be called echoes, and some additional ones besides. As a side note, observe that the interrogative versions in (113a) and (114a) are awkward, though not entirely unacceptable. This is unusual, and it will be commented on further in Section 4.3. As a general rule any rising declarative question can be replaced with its rising interrogative counterpart.

Examples of rising declaratives that question presuppositions appear in (47) (repeated above) and in (115) below.

(115) A: Maria's husband was at the party.

B's response:

- a. Is Maria married?
- b. Maria's married?
- c. #Maria's married.

(116) A (between bites): This persimmon is delicious!

B:

- a. Is that a persimmon?
- b. That's a persimmon?
- c. #That's a persimmon.

The presupposition need not be associated with a statement by the Addressee; a preceding question works too. The rising declarative in this case serves to indicate that the presupposition was in fact not shared information:

(117) A: Is the king of France bald?

B's response:

- a. Is France a monarchy?
- b. France is a monarchy?
- c. #France is a monarchy.

(118) A: What was Maria's husband wearing?

B's response:

- a. Is Maria married?
- b. Maria's married?
- c. #Maria's married.

The Speaker can also use a rising declarative to present an inference interpretable as a consequence of the Addressee's position, as shown in (119)-(121). In these examples all three locutions are acceptable, and all suggest that the A's preceding speech act has led the Speaker to the hypothesis or conclusion expressed by the descriptive content of the sentence.

(119) A: Jon has to leave early.

B's response:

- a. Will he miss the party then?
- b. He'll miss the party then?
- c. He'll miss the party then.

(120) A: Mark and Helena are leaving for Japan this week.

B: Oh...

- a. Did you talk to Helena?
- b. You talked to Helena?
- c. You talked to Helena.

(121) A to caller: Mom, I'll call you back tomorrow, OK?

Caller:

- a. Are you too busy to talk to your mother?
- b. You're too busy to talk to your mother?
- c. (I see.) You're too busy to talk to your mother.

More examples of the type shown in (119b)-(121b) can be found in Bartels 1997 and Noh 1998, who also make the point that sentences functioning as 'echoes' are not limited to repeating previously uttered content.

The contexts and inferential questions illustrated in (119)-(121) do not straightforwardly comply with the Contextual Bias Condition. Thus, more needs to

be said about why the examples are felicitous. The problem stems from the fact that in the framework employed here, the only readily available notion of ‘consequence’ is that of logical entailment in terms of possible worlds. This is a shortcoming in two ways. First, given the representation of individual commitment sets in terms of sets of worlds, it follows that individuals are committed to any logical entailments of their positions. This feature of the framework – that individuals are committed to any logical entailments of their explicit commitments – is of course at odds with reality. Individuals as a rule don’t realize all the consequences of their own and others’ commitments, though it is surely reasonable to expect that they recognize themselves as committed to at least some of them. About this problematic aspect of the framework I have nothing to say, beyond the disclaimer that the present proposal neither introduces nor relies upon it (and thus ought to be compatible with any system that offers a reasonable alternative).

The problem posed by these examples is a different, though related, one: the propositions expressed by the Speaker in (119)-(121) do not constitute logical entailments of the sentence uttered previously by the Addressee, or more generally, are not logically entailed by the Addressee’s public commitments. And entailment by the Addressee’s public commitment set is what the Contextual Bias Condition formally requires. For example, in (120), we would not want to say that if Mark and Helena are leaving for Japan this week, it has to be true (is logically entailed) that the Addressee talked to Helena. So the Contextual Bias Condition is not satisfied, and strictly speaking, we expect the declarative to be infelicitous, contrary

to the actual result. Generalized or conversational implicature is another possible route to inclusion in an individual's commitment set. But the propositions expressed by the Speaker in the above cases are not even (necessarily) implicatures, given that the preceding sentence may not ordinarily be taken to imply what the Speaker has chosen to represent as mutually inferrable from its utterance.

The ideal solution would be to state the Contextual Bias Condition in terms of a relation weaker than logical entailment, and/or adopt a richer representation of the context that allows for mutual inferrability as well as mutual belief. In other words, as already suggested, the ideal framework would incorporate a more realistic notion of consequence. Developing and/or defending such a notion is no small project, however. And for the purposes of working through the present hypotheses about questioning, whose overall architecture does not depend crucially on the particular definition of consequence adopted, it is not necessary to embark on that project. Instead I will make the simplifying assumption that (119)-(121) represent cases where the rising declaratives are *accommodated* as questions by making the necessary contextual adjustment to meet the Contextual Bias Condition (in the sense of *accommodation* originating with Lewis 1979.) The contextual adjustment required is the portrayal of the proposition expressed by the declarative as following from the Addressee's commitments. If we take *p* to be the content of the declarative question, what must be accommodated as a joint commitment of the

participants is $q \rightarrow p$, where q is a relevant public commitment of the Addressee's that serves as the basis for the inference.

In (121), for instance, A can be taken to conversationally implicate that he intends to cut short the conversation with his mother at the present time. His mother's response assumes that the reason for this intended action is that A is too busy to talk to her. In the declarative version (by hypothesis), she presents this assumption as a public commitment of A's. The form of $q \rightarrow p$ that must be accommodated is as follows:

(122) If *A intends to cut short the conversation with his mother* (q), then *he is too busy to talk to her* (p).

The somewhat aggrieved air of the mother's response is shared by all three versions and has to do, we may assume, with the descriptive content being presented rather than the locution type.

The possibility of accommodating a declarative as a question in contexts where the Contextual Bias Condition is not otherwise met is likely to be affected by other factors that help the Addressee determine the Speaker's intent. In particular, the relative knowledgeability of the two participants can play an important role. Consider the following example from Beun 2000. (123) presents an English translation from a corpus of Dutch dialogues involving an informant from the Amsterdam airport (A) and an information-seeker (B).

(123) A: Schiphol Information

B: Hello, this is G.M. I have to go to Helsinki, from Amsterdam. Can you tell me which flights leave next Sunday?

A: Just a moment.

A: Yes, there are several flights. One leaves at 9.10, one at 11.10, and one at 17.30.

B: The flight takes about three hours?⁹

In (123), it certainly does not follow from anything A says about departure times that the flight from Helsinki from Amsterdam takes about three hours. But given the situation as described, where A is mutually recognized as an official source of information about flights and B is in the role of information-seeker, it is pragmatically very unlikely that B would be *telling* A something about the flight duration. The declarative question in (123) is accommodatable, I suggest, due to the mutual premise inherent in the situation that A is informed as to facts about air travel from Schiphol and B is (relatively) not. (Imagine, for contrast, *The flight takes about three hours?* being uttered by A to B instead of the other way around. If such an utterance is interpretable, it is not as a question.)

The approach suggested above for accommodation of declarative questions is rather difficult to apply to (123). There is no particular commitment of the Addressee's from which the content of the declarative is taken to follow. Rather, there is a kind of blanket accommodation available for any declarative content presented by B that pertains to A's acknowledged area of expertise, i.e., airport

⁹ The question mark does not necessarily indicate rising intonation in Beun's examples, but just indicates that the declarative functions as a question. The discussion of (123) applies equally to rising and falling declarative questions, except where noted.

information. A general schema that allows accommodation of declarative questions in situations like that of (123) is given below:

- (124) a. A is mutually understood to be possessed of facts about some particular domain (in (123), airport operations).
b. B has reason to believe that some proposition *p* is a fact and that A knows it by virtue of (124a). (E.g., for (123) B believes *p* to be a fact about airport operations.)
c. Therefore, when B presents *p* to A declaratively, it can be taken to follow from *mutual* assumptions that A already knows, or is in a position to confirm, *p* – thereby providing the contextual adjustment necessary to satisfy the Contextual Bias Condition.

The crucial difference between the line of reasoning presented in (124), which leads to successful accommodation of the declarative as a question, and the reasoning about examples such as (107) earlier in this chapter, which did not, is represented by (124a). It is a built-in feature of the situation in (123) that the Addressee is publicly presented as knowledgeable about a certain body of facts. Situations where the Speaker is merely able to infer (privately) that the Addressee is likely to be knowledgeable will not allow accommodation in the same way. (Once again, it is to be hoped that a more sophisticated model of discourse beliefs, assumptions, and inferences will be able to capture this sort of interaction without recourse to the notion of accommodation.)

It does not follow from (124) that declarative questions can be used freely at any time in contexts where the Addressee is presumed knowledgeable about a particular domain. The limiting factor is expressed by (124b), which stipulates that the Speaker must have some reason (not necessarily public in this case) for believing the content of the declarative to be true, i.e., factual. This limitation is due

to the biasing potential of declaratives, as follows. The public assumption about the Addressee's *general* knowledgeability does not provide any clue as to the truth or falsehood of a *particular* claim – it's just assumed that the Addressee can provide the correct verdict if asked. Therefore, a Speaker who has no inkling as to whether a particular proposition is true can be expected to avoid offering that proposition in the form of a declarative question. The reason is that the declarative will bias the context toward its content, a move which (from the Speaker's point of view) is not motivated and has a 50/50 chance of requiring subsequent contextual repair. Conversely, if the Speaker *does* offer a declarative question in such circumstances, it may safely be inferred that the move *is* motivated – the Speaker has some reason to think that the Addressee will ratify the content. If the declarative is accommodated as a question in such circumstances, it will be understood that the Speaker has some (independent) reason to think that the content is true. This inference will go through even for rising declarative questions, which do not ordinarily commit the Speaker.

The prediction, then, is that doubtful or skeptical interpretations of rising declarative questions will be unavailable in contexts of this sort. And this prediction is correct. Consider the oddness of substituting (125) for B's question in the context of (123):

(125) B: The flight takes about three hours? #I thought it was shorter than that.

In light of this limitation, we might think of the subtype of declarative questions exemplified by (123) as *verification* questions – those that convey not

just the assumption that the Addressee will agree but the Speaker's belief that the proposition presented for approval is true. For contexts where considerations of relative knowledgeability dictate that a declarative question can only be interpreted as verifying, the difference between rising and falling declaratives is to a large extent neutralized. Beun notes that intonation was not crucial for questioning in the corpus studied; only 48% of the declaratives classified as questions in the corpus were rising.

Moving on to the next category of examples, the clearest evidence that rising declaratives are not inherently echoes comes from contexts in which there is no preceding utterance to echo at all. Consider (126)-(127). Here the declarative questions are infelicitous – a reasonable result, seemingly, since in the situations as described there is no reason for the Speaker to think that the Addressee is committed to the proposition expressed. In particular, there is no preceding utterance by the Addressee, as there has been in all the felicitous cases so far.

(126) Robin is sitting in a windowless computer room with no information about current weather conditions when another person enters.

Robin says to the newcomer:

- a. Is it raining?
- b. #It's raining?
- c. #It's raining.

(127) Laura and Max have just left a movie and are discussing it. Laura interjects:

- a. Are you hungry? (Let's get something to eat.)
- b. #You're hungry?
- c. #You're hungry.

But the absence of an appropriate utterance cannot be the decisive factor in the infelicity of the declaratives in (126)-(127), as can be demonstrated by altering the

non-linguistic context. In (128)-(129), the circumstances are altered slightly and the declaratives are felicitous. In the altered situations there *is* reason to think that the Addressee is committed to *p*, and the declaratives are accordingly improved. The pertinent evidence, however, is extra-linguistic – the wet raincoat in (128), the noisy stomach in (129).

[cf. (126)]

(128) Robin is sitting, as before, in a windowless computer room when another person enters. The newcomer is wearing a wet raincoat and boots. Robin says:

- a. Is it raining?
- b. It's raining?
- c. (I see that/So) It's raining.

[cf. (127)]

(129) In the middle of Laura and Max's discussion, Max's stomach rumbles audibly, providing evidence of his hunger. Laura interjects:

- a. Are you hungry? Let's get something to eat.
- b. You're hungry? Let's get something to eat.
- c. You're hungry. Let's get something to eat.

The contrast between (126)-(127) and (128)-(129) is clear, and it establishes decisively that rising declarative questions do not require a *linguistic* antecedent.

How do (128b)-(129b) manage to satisfy the Contextual Bias Condition? The explanation goes like this. In (128)-(129), but not the earlier cases, the Speaker is provided with evidence that the proposition expressed by the declarative is in fact true. Furthermore, the evidence is public – it is accessible to the Addressee as well. The public nature of the evidence is important, but the evidence plays a different role for the Addressee than it does for the Speaker. The Addressee is already in a position to know whether or not the proposition is true and need not rely on the

public evidence to decide that. After all, in (128) the Addressee has just been outside and presumably knows whether it's raining without having to reason from the state of her raincoat. The same point holds for Max's hunger in (129). The crucial contribution made by the public evidence from the point of view of the Addressee is that it enables the Addressee to recognize that the Speaker is being intentionally uninformative. That is, the Addressee is able to recognize that the Speaker assumes that the Addressee knows that the proposition expressed is true. Moreover, the Speaker, knowing that the Addressee can see the basis of the inference that the proposition is true, can count on the Addressee recognizing the intention to be uninformative. From the Speaker's point of view, the Addressee can be regarded as publicly committed to the proposition expressed – the Speaker believes it to be true, believes the Addressee to believe it, and believes the fact of the Addressee's belief to be mutual knowledge (or at least accommodatable as such). As for the Addressee, the public evidence ensures that she is in a position to appreciate the Speaker's point of view, even if the inference turns out to be wrong and the proposition is *not* true (and hence not truly believed by the Addressee).

As with (123), the situations in (128)-(129) do not allow for a skeptical reading of the declarative question (or the interrogative, for that matter). The questions in these cases function as verification questions, in the sense suggested in connection with (123). The reason is that the crucial role played by the nonlinguistic evidence in these cases is to make it clear to the Addressee that the Speaker takes the proposition expressed to be true (and *therefore* a public belief of

the Addressee's, who is understood to be in a position to know). A general hypothesis is suggested by (123) together with (128)-(129): skeptical or noncommittal readings of questions are only available when an overt act of commitment (usually by linguistic means) by the Addressee to the proposition in question precedes the declarative utterance. With respect to (128)-(129) in particular, it is hard to imagine a case where there would be nonlinguistic evidence strong enough to warrant a public assumption by the Speaker that the Addressee is committed to the particular proposition expressed by a question while at the same time leaving open the possibility that the Speaker has not inferred from that same evidence that the proposition is true.

In case the above explanation of how (128)-(129) satisfy the Contextual Bias Condition seems unnecessarily complicated, let us confirm that the complications are indeed motivated. Consider once more the simpler possibility that what's required is that the Speaker have evidence for the Addressee's belief. In the echo question analysis of Noh 1998, which is similar to the analysis given here in that it involves attribution of propositional content to the Addressee, such a possibility is suggested. Noh cites examples similar to (128)-(129) in the course of arguing that so-called "echo questions" do not require a prior utterance and remarks on the contrast between the (a) and (b) cases in (130):

(130) [Noh's (15)]

A sees B walking towards the door, and says:

- a. You're off to catch the train? [also fine: *Are you off to catch the train?*]
- b. #Henry VIII had seven wives?
- c. Did Henry VIII have seven wives?

The explanation suggested by Noh for the pattern in (130) is as follows:

(a) is acceptable because A attributes to B a thought that B could plausibly be supposed to have; (b) is unacceptable because the thought attributed is one B could not plausibly be supposed to have in the circumstances; (c) is acceptable because it involves no attributed thought.

[Noh 1998, p. 607]

But this sort of explanation, to the extent it allows the plausibility of the attribution to be based on evidence accessible only to the Speaker, is not sufficient. Suppose the Speaker of (130a) believes (but desires confirmation) that Henry VIII had seven wives. Furthermore, the Speaker happens to know, unbeknownst to the Addressee, that the Addressee watched a documentary on Henry VIII the preceding evening (from which the Addressee presumably would have acquired the fact in question). Even though it is plausible for the Speaker to assume that the Addressee believes the proposition, the private justification does not particularly improve (130a).

Similarly, with reference to (126), suppose that Robin has access to information about the current weather conditions, unbeknownst to the newcomer. (Robin might, for example, have just spoken on the phone with someone knowledgeable or have accessed a web site with up-to-the-minute local reports.) Assuming that Robin has good reason to be biased herself, together with the assumption that the Addressee is knowledgeable and may be presumed to have the same bias, is not sufficient by itself to improve (126) (just as similar assumptions failed to improve (107b) earlier.) With reference to (127), Laura may have a perfectly good private reason for believing that Max is hungry. Perhaps she has noticed, though he has not, that he's invariably hungry after watching a Woody

Allen movie. But again, the mere existence of a private bias on Laura's part, no matter how well-founded, doesn't regularize the rising declarative question.

In short, the facts show that no amount of tinkering with assumptions about private knowledge, private belief, or private evidence will render a declarative question felicitous in the absence of relevant *public* evidence. In its most obvious manifestation, the public evidence for the Addressee's belief is the Addressee's own utterance. But as (128)-(129) show, the evidence need not be linguistic in nature, as long as the Contextual Bias Condition can be met (or accommodated). The result arrived at empirically in this section is thus in agreement with the hypothesis advanced on more conceptual grounds in Section 4.1.

Throughout the examples in this section we can see once again the significant generalization pointed out at the end of Section 2.3: generally speaking, wherever a rising declarative is acceptable as a question, the corresponding interrogative is as well; and the effect produced is very similar. It is noteworthy, in particular, that interrogatives can be understood as presenting an inference, as in (119a)-(121a).

4.3 Polar questions defined

In Sections 4.1 and 4.2 I defended the claim that only declaratives understood as uninformative with respect to the Addressee's commitments can be interpreted as questions. So far, however, I have not said much about what a "question" is, instead relying implicitly on our strong intuitions about rising declaratives in

various contexts. It is now time to ask what underlies those intuitions of questionhood. The rising declarative questions exemplified throughout this paper, as well as the interrogatives that accompany them, are compatible with a variety of attitudes, intentions, and discourse effects. In some uses, these questions seem to fit the paradigm of requesting a yes/no response from the Addressee, with the Speaker motivated by a desire to confirm that the proposition in question is true. But in others, for example the expression of doubt exemplified in (44), the rising declarative question seems to be used in a more expressive way to register the Speaker's reaction, with the response of the Addressee being a secondary consideration. Furthermore, given the Contextual Bias Condition, the Addressee's expected response to a rising declarative question is already available in the context, if not already stated. All of these considerations might lead us to wonder in what sense rising declaratives constitute questions at all. Nevertheless, it is perfectly natural and commonplace to describe them that way in the uses we have seen.

In considering declaratives as questions in the abstract, attention naturally gravitates toward rising declaratives; falling declaratives are more readily associated in isolation with assertive uses. But as corpus studies testify (e.g., Geluykens 1988, Beun 2000), falling declaratives are a relatively common way to ask a question in the functional sense of soliciting a yes-no response from a knowledgeable addressee. And this ability of falling declaratives to function as 'questions' in some sense must also be taken into account.

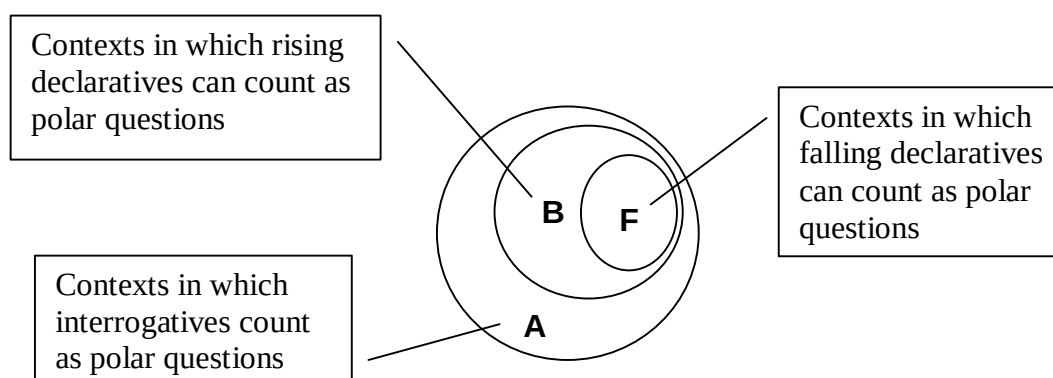
The challenge posed by the above observations is characterizing the notion of polar question in such a way as to admit a variety of locution types and intended uses while also keeping hold of our intuitions about relative ‘naturalness’ as questions. The key to this puzzle in the present account lies in the distributional pattern of interrogatives vis-à-vis declarative questions, together with consideration of the context update meaning associated with each locution type, as defined in Chapter 3.

With reference first to rising declaratives and interrogatives, recall that rising declaratives, in contexts where they function in ways we are inclined to call *questioning*, seem to be interpretable in the same ways as their interrogative counterparts. This was pointed out in Sections 2.3 and 4.2, where it was also noted that interrogatives can generally be used wherever rising declaratives are felicitous as questions. The solution suggested by this distributional pattern is a simple one: rising declaratives count as polar questions when, and because, their effect on the context is the same as that of a polar interrogative with the same content. As was remarked in Section 4.1, there is a general sense in which we are inclined to regard any (felicitous) use of a polar interrogative as a question, regardless of what it is used to accomplish in a particular context. Accepting this general sense as the appropriate one, understanding why rising declaratives function naturally as questions is a matter of understanding their overlap with interrogatives. This we are now in a good position to do.

In similar fashion, the contrast between rising and falling declaratives as questions can be characterized in distributional terms: the distribution and effects of falling declaratives do not overlap with that of interrogatives to the same extent as rising declaratives. As we have already seen, there is a crucial difference between rising and falling declaratives – the latter, but not the former, result in Speaker commitment to their content. And this Speaker commitment, I will show, restricts both the domain and the range of falling declaratives used as questions relative to their rising declarative and interrogative counterparts. Although rising declaratives also express commitment, on the part of the Addressee, this commitment is already required by the Contextual Bias Condition and thus does not result in *additional* restrictions the way that Speaker commitment does for falling declaratives.

The distributional picture that will emerge in this section is represented schematically in Figure 1:

Figure 1: Distribution of declarative questions



In Section 4.4 I will discuss the implications of this distributional picture in more detail and argue that its accuracy provides empirical support for the treatment of polar questions advocated here. For now, it serves to preview the discussion in the remainder of this section, in which I explicitly define the notion of ‘polar question’ and attempt to make concrete our intuitions about rising vs. falling declaratives as questions.

The proposal that every felicitous utterance of a (rising) polar interrogative counts as a polar question is expressed by the sufficiency condition in (131):

Sufficiency condition

(131) An utterance of $\uparrow S_{\text{interr}}$ is a polar question in C if C admits $\uparrow S_{\text{interr}}$.

To allow utterances of other locution types to qualify as polar questions, we need a definition that accomplishes what (131) does but that also incorporates the necessary restrictions on declarative questions. There is an obvious candidate for this role – the Uninformativeness Condition introduced in Section 4.1. That condition is repeated in (132), this time as a necessary *and* sufficient condition for polar questionhood. (It will be revised later in this section in light of additional restrictions applying to falling declarative questions.)

Definition of polar question (first version)

(132) An utterance of L is a polar question in C iff L is uninformative with respect to $CS_{\text{Addr}}(C)$.

[Uninformativeness Condition, adapted from (104)]

(132) is a perfectly adequate definition as far as rising declaratives and interrogatives go. Every felicitous utterance of a rising interrogative will satisfy (132), as desired, because interrogatives are uninformative by definition. Every felicitous utterance of a rising declarative in a context that meets the Contextual Bias Condition will qualify as a polar question under (132), since a rising declarative is uninformative with respect to cs_{Addr} just in case its content is already a public commitment of the Addressee. Rising declaratives do not qualify as questions in other contexts. That is the result we are seeking.

Note that (132) does not exclude utterances of locutions that are *vacuous* with respect to the context, i.e., locutions whose content is already a joint commitment. (See Section 3.5, p. 62.) I will assume that in general, vacuous questions, like other vacuous moves, are unexpected, and anomalous to the extent they do occur. This general assumption frees me from having to stipulate their exclusion in each definition of utterance categories.

(132) correctly predicts that contexts admitting rising declaratives as questions can also be expected to admit the corresponding interrogatives, a point to be discussed in the next section. With that prediction, we have in place the first piece of the story of the overlap between rising declaratives and interrogatives: the set of contexts in which rising declaratives occur as polar questions is a proper subset of the set of contexts in which interrogatives occur as polar questions.

The second crucial aspect of the affinity between rising declaratives and interrogatives has to do with their effects in contexts where both are possible.

Under the present proposal, rising declaratives and interrogatives are identical in effect in such contexts, as (133) states:

(133) If an utterance of $\uparrow S_{\text{decl}}$ is a polar question in C , $\uparrow S_{\text{decl}}(C) = \uparrow S_{\text{interr}}(C) = C$,
where $\uparrow S_{\text{interr}}$ and $\uparrow S_{\text{decl}}$ have the same descriptive content.

The equivalence holds because in any context where a rising declarative is uninformative with respect to the Addressee's commitments, as required for (132), it is also entailed by the context as a whole. And interrogatives are entailed by any context that admits them. The definition of entailment is repeated below for reference:

(94) $C \models L$ iff C admits L and $C + L = C$

This formal correspondence between the effects of rising declaratives and interrogatives is inherent to these particular locution types, in the sense that it follows solely from the definition of locution meanings given in Chapter 3. That is, identity of effect with rising interrogatives in certain contexts is a built-in feature of rising declaratives (and *vice versa*). The claim I make here is simply that rising declaratives are 'natural' as questions for the following reason: in a context where a rising declarative is uninformative with respect to the Addressee's commitments, it follows *by virtue of the locution meaning* that it mimics the effect of an interrogative in the same context. Falling declaratives differ on this point, as we will see shortly.

The correspondence between rising declaratives and interrogatives is as much a fact about the nature of interrogatives in English as it is about declaratives. It is because rising interrogatives can be used felicitously in contexts where the Addressee is already committed that the overlap in contexts and effects between different sentence types is possible.

It will be useful to have a name for the subcategory of polar questions whose characteristic contextual feature is the prior commitment of the Addressee. I will call these *reiterative questions*, as defined in (134):

Reiterative questions

(134) An utterance of L with descriptive content p is a *reiterative question* in C iff:

- a. The utterance of L is a polar question in C.
- b. p is a commitment of the Addressee in C.

All utterances of rising declaratives that qualify as polar questions are also reiterative questions. Some, but not all, utterances of rising interrogatives are reiterative questions.

One more point needs to be made about the effects of rising declaratives and interrogatives before we move on to falling declaratives. It will be recalled that the present analysis of interrogatives, under which a rising interrogative represents simply an identity function on a commitment set, was identified at the outset as a partial and admittedly inadequate characterization of the effects of interrogatives on the context. The concern is that when a more complete story is told about interrogatives, the correspondence between the effects of the two sentence types in

certain contexts will reveal itself to be an artifact of the restricted account, and will vanish along with the explanation of questioning.

It is difficult to argue this point conclusively without reference to a particular analysis of interrogatives. But there is a general consideration which suggests that the explanation of rising declarative questions in terms of the effects of interrogatives will survive under a more complete analysis. Rising declarative utterances, in contexts where they qualify as polar questions, do not change any commitment sets. This really constitutes their crucial point of resemblance with interrogatives. In a more complete account, interrogatives might be defined to do *more* by operating on other aspects of the context, but in this case it will still hold that a rising declarative polar question will share with an interrogative the property of not altering commitment sets. The generalization expressed in (133) can easily be restated to capture that if need be.

Turning now to falling declaratives, we find a somewhat different picture. If we leave (132) as it stands, falling declaratives ought to qualify as polar questions in the same set of contexts that rising declaratives do, namely in any context where the Contextual Bias Condition is met. But that is not the correct result for falling declaratives. We have seen a number of examples where a rising declarative operates as a question but the corresponding falling declarative is unacceptable (or less acceptable). There are three kinds of cases to worry about, illustrated by the examples repeated below:

- (44) [A&B are looking at a co-worker's much-dented car.]
 A: His driving has gotten a lot better.
 B's response:
 a. Has it? I don't see much evidence of that.
 b. It has? I don't see much evidence of that.
 c. It has. #I don't see much evidence of that.
- (118) A: Maria's husband was at the party.
 B's response:
 a. Is Maria married?
 b. Maria's married?
 c. #Maria's married.
- (128) Robin is sitting, as before, in a windowless computer room when another person enters. The newcomer is wearing a wet raincoat and boots. Robin says:
 a. Is it raining?
 b. It's raining?
 c. (I see that/So) It's raining.

In connection with (44), the proposal that falling declaratives commit the Speaker to their content has already given us an explanation of the oddness of (44c). The follow-up remark suggests skepticism or at least reservations on the Speaker's part, an attitude that conflicts with the commitment expressed by the preceding locution. The importance of (44) at present is that it illustrates the reduced range of falling declaratives with respect to the Speaker's attitude. Unlike rising declaratives and interrogatives, which allow for a spectrum of Speaker attitudes, falling declaratives limit the Speaker's future options with respect to the content expressed – that is the essence of Speaker commitment. Given that the Contextual Bias Condition (or Uninformativeness Condition) also requires that the Addressee have a prior commitment to the content of the declarative, the characteristic effect of a falling

declarative as a question¹⁰ will be to make the content of the declarative a joint commitment in the context. I refer to this sort of question as *resolving* and define it in (135):

- (135) An utterance of L with descriptive content p is a *resolving question* in C iff:
- a. The utterance is a reiterative question in C.
 - b. p is unresolved in C and a joint commitment in C + L.

Only falling declaratives can qualify as resolving questions.¹¹ Neither rising declaratives nor interrogatives have the capacity to commit the Speaker to their content, which is what (135) requires. Since qualifying as a reiterative question means that the Addressee is already committed to p in C, the requirement of (135b) that p be unresolved in C can only be satisfied if the Speaker starts out uncommitted, then becomes committed as the result of the move.

We thus have the first piece of an explanation as to why falling declaratives are less natural as questions than rising ones. Recall the generalization stated earlier about the equivalence of effects between rising declarative questions and interrogatives:

¹⁰ It might be argued that falling declaratives do not constitute questions at all, at least not in the same sense that rising declaratives and interrogatives do. If (133) were adopted as a necessary condition for questionhood, instead of being stated as an auxiliary generalization, falling declaratives would effectively be ruled out as polar questions. This is a terminological matter, not a substantive one. I certainly do not deny that falling declaratives can be used in question-like ways in discourse. The issue at stake is how to characterize 'question-like' and whether falling declaratives should be included in the same category as rising declarative and interrogative questions or receive their own category. The decision either way is not critical, since the category of *polar question* plays an explanatory role but has no formal significance in the proposal. I choose to allow falling declaratives to qualify as a variety of polar question, distinct in effect from rising declaratives and interrogatives.

¹¹ (135) is rather broad and may include instances of falling declaratives that are simply used to express agreement. It seems there may be a further requirement for falling declaratives to be interpreted as questions, namely that the Speaker actually pause and turn over the conversational controls to the Addressee. I will not worry about this point, however, as the distributional generalizations that are the point of this section hold even if the category of falling declarative questions is too broadly defined.

(133) If an utterance of $\uparrow S_{\text{decl}}$ is a polar question in C , $\uparrow S_{\text{decl}}(C) = \uparrow S_{\text{interr}}(C) = C$, where $\uparrow S_{\text{interr}}$ and $\uparrow S_{\text{decl}}$ have the same descriptive content.

No such equivalence holds for falling declaratives as questions. The only way that a falling declarative question can ever be equivalent in effect to a rising interrogative is if it is vacuous with respect to the context, and vacuous moves, as already suggested, are not very significant. So falling declarative questions crucially differ from rising declarative ones in not having the same contextual effect as rising interrogatives in the same contexts.

Turning to the type of presuppositional question exemplified in (118), it is immediately evident that a different explanation must be sought for the oddness of (118c). Here the problem cannot be contradictoriness, as it was for (44c), since the content of the falling declarative conflicts neither with a preceding statement by the Speaker nor with the Addressee's preceding statement, which indeed presupposes the content of (118c).

Before attempting to account for the oddness of (118c), I will first deal with the third category of example, repeated below:

- (128) Robin is sitting, as before, in a windowless computer room when another person enters. The newcomer is wearing a wet raincoat and boots. Robin says:
- a. Is it raining?
 - b. It's raining?
 - c. (I see that/So) It's raining.

In (128), the falling declarative does seem to work as a question in the broad sense of being interpretable as an appeal for a yes/no response, and it qualifies as a resolving question under (135) (given that all of (128a-c) can be taken to be

reiterative questions). But (128c) works much better when one of the parenthesized additions is present, making it clear that the Speaker has just inferred the proposition presented by the declarative. This type of augmentation is never required for a rising declarative or an interrogative (and is not always necessary with a falling declarative). It suggests the falling declarative as a question is aided by additional contextual support of a particular sort. (Lindsey 1985:83 also notes that so enhances the interpretation of a falling declarative as a question.)

This additional contextual support has to do, I propose, with the basis for the Speaker's commitment. In the case of a questioning use of a falling declarative, it must be clear contextually that the Speaker's commitment is not motivated independently (i.e., by the Speaker's own private evidence) but is contingent upon the Addressee's commitment. In effect, it must be understood that the Speaker is committing to the content of the declarative with the caveat that the Addressee is more authoritative (or at least *as* authoritative) as the Speaker with respect to the matter in question. Inferential expressions such as *I see* in (128c) help to produce the required effect because they make it clear that the Speaker has concluded that it is raining out based on the evidence on hand, rather than having some private and independent source of knowledge. With regard to the subject of current weather conditions, the Addressee is undoubtedly in a better position than the Speaker in (128), and the mutually understood superiority of the Addressee's position facilitates interpretation of the Speaker's utterance as a question.

To capture this additional nuance associated with falling declarative questions, I will add a clause to the definition of polar question:

Polar question (final version)

- (136) An utterance of L with descriptive content p is a polar question in C iff (a) and (b) hold:
- a. L is uninformative with respect to $cs_{Addr}(C)$.
 - b. If $cs_{Spkr}(C + L) \subseteq p$, the Speaker's commitment to p is mutually understood as contingent upon the Addressee's commitment to p.

Clause (136b) as stated is quite general, applying to any utterance that either accomplishes the task of committing the Speaker to p or that takes place in a context where that commitment has already been accomplished. As we know from the definition of falling declarative locution meaning, (136b) will always apply to an utterance of a falling declarative, since falling declaratives always result in a context where the Speaker is committed to p. This ensures that the additional restriction is always in effect for falling declaratives as questions, as desired. There is no requirement that a falling declarative must be accompanied by any particular indicator; (136b) just specifies what must be inferrable in some manner from the context, which includes the possibility of linguistic aids such as *I see*, *so*, *oh*, etc. as well as extra-linguistic clues.

For rising declarative questions, the addition of (136b) makes no substantive difference. Rising declaratives fail to commit the Speaker to p, by definition. So the only way clause (136b) can come into play for a rising declarative utterance is if the locution is vacuous – i.e., if the Addressee is already committed to p in C, satisfying (136a), and the Speaker is as well, satisfying the antecedent in (136b).

Vacuous moves, I have suggested, are to be ruled out independently on general principles. In any case, the requirement in (136b) does no harm.

As for rising interrogatives, they need not be vacuous for (136b) to take effect. If the Addressee is not already committed to *p* in *C* but the Speaker is, then the requirement in (136b) will be in force. I believe this to be a reasonable result, even though interrogatives don't seem to exhibit additional contextual requirements of the sort that falling declarative questions do. What I will assume is that a rising interrogative actually conveys, as part of its semantic contribution, the understanding that the Speaker is prepared to commit to *p* only if the Addressee does. In the kind of contexts relevant for (136b), where the Speaker is committed to *p*, this component of interrogative meaning will ensure that the Speaker's commitment is viewed as contingent upon the Addressee's, as required. In other words, I assume that an interrogative will always satisfy (136b), by definition. This proposal has the same shape as the claim made in Section 4.1 about uninformativeness with respect to the Addressee's commitments – there is a property associated with questioning that interrogatives have by nature, but that declaratives must acquire by means of the context. I will not attempt to justify this assumption about interrogatives here, instead leaving it in its speculative state, as

motivating and modeling this proposed aspect of interrogative meaning is beyond the scope of the present work.¹²

With the addition of clause (136b) to the definition of polar question, the second piece of the explanation for the contrast between falling and rising declaratives as questions is now in place. The fact that contexts allowing falling declaratives as questions are subject to the additional restriction represented by (136b), effectively above and beyond the requirements for rising declarative questions, means that falling declaratives are possible as questions in a proper subset of the contexts supporting rising declaratives as questions. To use a familiar linguistic term, falling declaratives are more *marked* as polar questions than rising declaratives. Here markedness can be defined in terms of distribution across contexts, as follows:

- (137) Given the definition of an utterance category UC and locutions L and L', L is *more marked* than L' as an instance of UC iff $\{ C: \text{an utterance of L qualifies as UC in } C \} \subset \{ C: \text{an utterance of L' qualifies as UC in } C \}$.

The generalizations about distribution of the three locution types as polar questions can then be stated as in (138)-(139):

- (138) A rising declarative is more marked as a polar question than a rising interrogative with the same content.
(139) A falling declarative is more marked as a polar question than a rising declarative (or rising interrogative) with the same content.

¹² Some rudimentary support is provided by the nuances of the shoplifting-type examples discussed in Section 2.3, where the interrogative was observed to carry the insinuation that the Addressee is in a position to pass judgment on the content presented.

Note that this notion of markedness across contexts is distinct from the relative frequency of utterances of certain types in discourse.

In summary, then, the definition in (136) stands as a workable characterization of the pragmatic category of polar questions. Armed with (136) and the considerations that motivated it, let us now revisit the final example of an unacceptable falling declarative question to be accounted for, repeated below:

- (118) A: Maria's husband was at the party.
B's response:
a. Is Maria married?
b. Maria's married?
c. #Maria's married.

The point to be made about (118) is that it is not as distinct from felicitous declarative questions like (128) as it initially appeared to be. When we add supporting material of the sort that is helpful in satisfying (136b), (118c) also improves:

- (140) A: Maria's husband was at the party.
B's response:
a. Is Maria married?
b. Maria's married?
c. Oh, so Maria's married.

Here the supporting material seems to be absolutely essential for the felicity of the falling declarative question. To the extent that the falling declarative becomes felicitous with this addition, it patterns with the rising declarative and interrogative in conveying that the 'presupposition' it expresses was not in fact already a joint commitment, as the Addressee had assumed. Rather, like its rising declarative and interrogative counterparts in this situation, the falling declarative question suggests

that its content is news as far as the Speaker is concerned. (I will discuss this phenomenon further in Section 4.5.) The important point at the moment is that the same effect obtains for all three locutions – the falling declarative is not anomalous in this respect, provided the appropriate supporting material is present. Unlike rising declaratives and interrogatives, however, the falling declarative version also expresses the Speaker’s commitment, and is thus incompatible with the readings of skepticism or doubt that are readily available for the first two locution types in the same context.

4.4 The distribution of rising declarative questions revisited

In this section I return to points made earlier about the distribution and interpretation of declarative questions, in particular rising declarative questions, and interrogatives. I will argue that the distributional facts support the strategy adopted in the previous section of understanding the questioning nature of rising declaratives in terms of their overlap in effect with rising interrogatives.

The claim that the sense in which rising declaratives function as questions is derived from, and nondistinct from, the sense in which interrogatives function as questions has the virtue of explaining certain otherwise mysterious gaps in the distribution of rising declarative questions. So far we have seen two relevant classes of contexts:

- (141) Contexts in which a rising interrogative is acceptable but the corresponding rising declarative is unacceptable as a question.

(142) Contexts in which both the rising interrogative and the rising declarative versions of a question are acceptable.

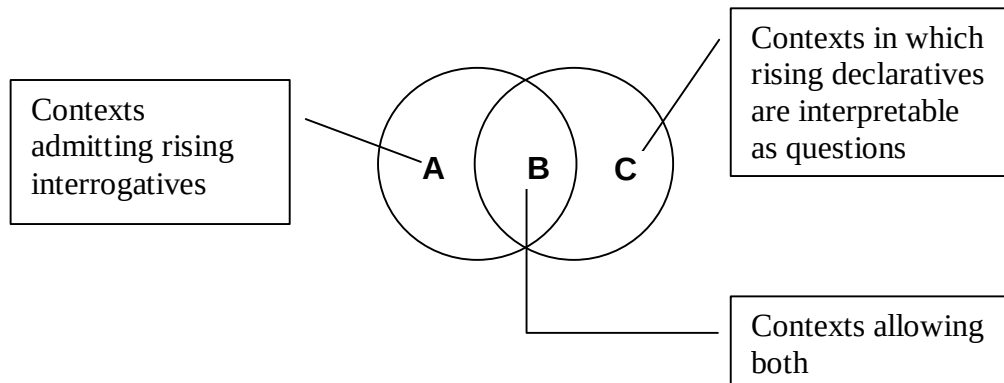
The contexts described by (141) were exemplified in Section 2.2 and some of the examples in Section 4.2 (e.g., (107), (126)). The second type, in which both versions are fine, can be seen in Section 2.3 and throughout Section 4.2 (e.g., (115), (119), (128)).

Suppose now that, contrary to the hypothesis advanced above, rising declarative questions are assumed to constitute a distinct variety of questions, differing essentially in at least some of their uses from questions asked via interrogatives. (This is not an implausible supposition on its face given that there are undoubtedly some important differences between interrogative and declarative questions.) Then alongside the types of contexts described in (141)-(142) we might expect to find a third type in which *only* rising declarative questions are possible:

(143) Contexts in which rising declaratives are acceptable as questions and rising interrogatives are not.

Schematically, the picture would be as in Figure 2:

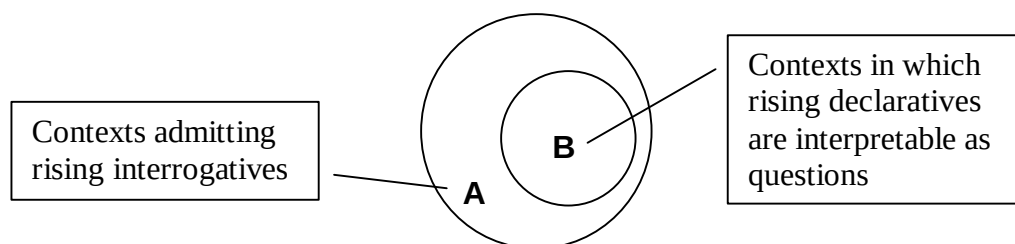
Figure 2: Hypothetical distribution of rising declarative questions



In fact we have seen few examples of C-type contexts so far, and for good reason. Contexts of this sort, i.e., contexts in which rising declaratives can be used as questions but rising interrogatives cannot be, are rare to non-existent.

The distribution of rising declarative questions, as was anticipated by Figure 1 above, is more accurately schematized as in Figure 3:

Figure 3: Actual distribution of rising declarative questions



The current proposal predicts the absence of C-type contexts and thus is in accord with the distribution represented in Figure 3. The distributional gaps follow from

the simple expedient adopted here of defining questionhood in terms of the distribution of interrogatives.

The picture presented in Figure 3 oversimplifies slightly in ignoring two systematic classes of exceptions to the generalization that rising interrogatives are acceptable wherever rising declaratives are interpretable as questions. First, interrogatives are awkward as ‘strict’ echoes compared to rising declaratives.

Examples from Section 4.2 are repeated in illustration:

(113) A: There’s a leopard in the living room.

B’s response:

- a. ? Is there a leopard in the living room?
- b. There’s a leopard in the living room?
- c. There’s a leopard in the living room.

(114) A: Gina went skydiving yesterday.

B: You’re kidding!

- a. ? Did she jump out of an airplane?
- b. She jumped out of an airplane?
- c. #She jumped out of an airplane.

It seems too strong to call the interrogative fully infelicitous in (113a) and (114a), but it is somewhat degraded compared to (113b)-(114b). I use the term ‘strict echo’ informally to distinguish a case like (113) from the more elliptical repetitions illustrated in Section 2.3 and in (144) below, where the awkwardness of the interrogative disappears.

(144) A: There’s a leopard in the living room.

B’s response:

- a. Is there?
- b. There is?
- c. There is.

I assume that the important difference between ‘strict’ and elliptical echoing is that the former retains the focus structure of the original – the nuclear accent is placed so as to produce the same focusing effect. This is clear enough intuitively though not so easy to capture formally in light of examples like (114), where there is no lexical element shared with the original. It cannot simply be said that the focus must fall on the ‘same’ element; what’s important is that both (113b) and (114b) have broad, “all-new” focus. The strictly echoing utterance is offered *as if* it were new, when it manifestly is not. The ellipsis in an elliptical repetition, by contrast, depends on the elided information *not* being new.

Intuitively, the oddness of the interrogative in (113a)-(114a) is understandable, assuming that ‘strict’ echoing is metalinguistic in nature (see Noh 1998 on resemblance of linguistic form). With metalinguistic echoes, we expect reiteration of sentence type along with descriptive content. I will assume that some such explanation is available and will not be further concerned with this sort of counterexample here. Note that on the present account ‘echoing’ is taken to be a phenomenon defined by repetition of descriptive content (and possibly sentence type) and is not tied to any particular intonational pattern or discourse function. This characterization differs from that of, e.g., Bartels 1997, who agrees with Quirk et al. in taking the discourse function of questioning to be an essential component of an ‘echo’. Although this topic deserves more discussion, I won’t undertake it here.

The second kind of exception to the picture presented in Figure 3 involves the infelicity of certain sorts of modifiers with interrogatives, seen in Section 2.2 for examples like (33):

- (33) a. #Has the manager *of course* been informed?
b. The manager has *of course* been informed?
c. The manager has *of course* been informed.

The problem is that while the rising declaratives in these examples seem intuitively question-like, they violate the generalization of (104) because the corresponding interrogatives are infelicitous. A way out is suggested by Huddleston's remark in connection with (33) and related examples that bias markers (such as *of course*) are 'outside the propositional content of the question'. If this position can be sustained, the generalization can be preserved by stating it in terms of propositional content without the offending modifiers. Again, I will assume without further discussion that a resolution along these lines is feasible.

Returning to the distributional picture, observe that there is one more type of context, not delineated in Figure 2, whose absence is both significant and predicted by the approach taken here. If rising declarative questions were distinct in important ways from interrogative ones, we would not be surprised to find that interrogative and declarative questions sometimes had disparate effects in B-type contexts where both are possible. They might be expected to interact with the context in distinct ways, producing two different flavors of question in the same environment. If such a possibility were realized, there would be an identifiable subset of the B-type contexts which produce the divergent interpretations. But as

already mentioned, such contexts are unattested. As we have seen throughout the paper, in contexts where both are possible, the interpretations of rising declaratives and interrogatives converge. This is a striking fact, and it follows directly under the analysis of declarative questioning I have offered here.

4.5 *What reiterative questions are good for*

In this section I will survey briefly some of the discourse goals and propositional attitudes that declaratives used as questions are compatible with. To begin with, let me emphasize that I am *not* proposing a view where sentence form and/or intonational category associate directly with speech acts or discourse functions. To the contrary, I have argued at some length in the preceding sections of this chapter that the questioning function of declaratives is derived through the interaction of locution meaning and context, rather than being assigned as a primitive. Furthermore, the notion of ‘polar question’ developed above is an extremely general one that encompasses any number of discourse functions. Nor have sentence type and intonational category been associated with particular propositional attitudes of the Speaker and Addressee. The idea that declaratives express commitment was developed in Chapter 3 as a property of a type of update, not directly in terms of speech acts or attitudes.

Nevertheless, we do expect some sort of systematic relationship between the meaning of a locution and what it can be used to accomplish in a discourse, as well

as what it indicates about the Speaker's attitude and assumptions about the Addressee. The kind of relationship consistent with the present analysis is one in which the locution meaning – including the contributions of both sentence form and intonational category – operates to *constrain* what an utterance of that locution can be used to do without determining its function. In a way, the present analysis has more predictions to make about what declarative questions *can't* do than what they can.

For example, one of the observations with which this thesis began is that declaratives, rising and falling, are unacceptable as neutral questions, i.e., questions where no particular answer is anticipated. Now that we have a working definition of a polar question from Section 4.3, that generalization can be better understood. I define a *neutral question* as in (145):

Neutral question

(145) An utterance of L is a neutral question in C iff:

- a. L is a polar question in C.
- b. C + L is neutral with respect to the descriptive content of L.

A neutral question is one that is uttered in a neutral context and that preserves the neutrality of the context. It follows from the definitions of locution meaning in Section 3.2 that no declarative will ever qualify as a neutral question. (Recall theorem (91), p. 60, which stated that no declarative locution is ever neutral with respect to a context.) Only an interrogative can achieve the feat of preserving neutrality. Declarative meaning thus limits the functioning of declarative locutions as questions in a concrete way.

Declaratives as questions are characteristically *reiterative*; the definition is repeated below:

- (134) An utterance of L with descriptive content p is a *reiterative question* in C iff:
- a. The utterance of L is a polar question in C.
 - b. p is a commitment of the Addressee in C.

The issue that arises with respect to reiterative questions is why they should be useful for anything at all, given that the Addressee's expected response must, by definition, already be available to the Speaker. In the remainder of this section I will address that issue by discussing a few of the uses to which reiterative questions may be put, with reference to examples introduced earlier.

The first point to be made is that an utterance may qualify as a reiterative question without the Addressee ever having made an explicit, intentional commitment to the proposition in question. Recall examples like (128)-(129), in which there was no preceding utterance by the Addressee:

- (128) Robin is sitting, as before, in a windowless computer room when another person enters. The newcomer is wearing a wet raincoat and boots. Robin says:
- a. Is it raining?
 - b. It's raining?
 - c. (I see that/So) It's raining.
- (129) In the middle of Laura and Max's discussion, Max's stomach rumbles audibly, providing evidence of his hunger. Laura interjects:
- a. Are you hungry? Let's get something to eat.
 - b. You're hungry? Let's get something to eat.
 - c. You're hungry. Let's get something to eat.

The reiterative question in (128) has the effect of raising the issue of what the weather conditions are like. Even if the Addressee's answer is a foregone

conclusion, there are effects to be achieved by this move. In the terminology of Section 3.5 (p. 62), a reiterative question is not *vacuous* in this situation because the knowledge that it's raining is not (yet) mutual knowledge. The reiterative question serves the purpose of letting the Addressee know that the Speaker realizes that it's raining, and this is an essential step for the proposition becoming a joint commitment. Thus, if the Speaker's goal is to achieve consensus on the proposition expressed, a reiterative question is a reasonable move.

Of course, the Speaker's inference may always be wrong in cases like (128)-(129), and a further obvious purpose of the reiterative question is to check that inference. Although the Speaker in (128)-(129) has reason to believe that the proposition in question is true, there is no doubt that the Addressee is in a much better position to vouch for it. Using a reiterative question, particularly a declarative version, makes it crystal clear that the proposition will become a joint commitment (if it does) on the authority of the Addressee, not the Speaker.

There is some ambiguity as to whether the interrogatives in (128a)-(129a) are interpreted as reiterative questions at all. Since as just noted, the Addressee hasn't *explicitly* made a commitment in these cases, there is room for reading the interrogative as a neutral question. This possibility, which is not available for the declarative versions, accounts for the slightly different social consequences of uttering (129a), as opposed to (129b) or (129c). The declaratives rely for their effect on Max's recognition that Laura has noticed his stomach growling. By contrast, the interrogative version at least allows the participants to pretend that the

timing of Laura's question was coincidental, and hence can be viewed as less intrusive, more polite. Thus, use of an interrogative may exploit the ambiguity in a particular context as to whether the Addressee is committed or not in a way that is not available to declarative questions.

The points made above with respect to (128)-(129) also apply in large part to the subcategory of reiterative questions exemplified by (119)-(121), repeated below:

(119) A: Jon has to leave early.

B's response:

- a. Will he miss the party then?
- b. He'll miss the party then?
- c. He'll miss the party then.

(120) A: Mark and Helena are leaving for Japan this week.

B: Oh...

- a. Did you talk to Helena?
- b. You talked to Helena?
- c. You talked to Helena.

(121) A to caller: Mom, I'll call you back tomorrow, OK?

Caller:

- a. Are you too busy to talk to your mother?
- b. You're too busy to talk to your mother?
- c. (I see.) You're too busy to talk to your mother.

In these cases there is a preceding utterance by the Addressee, but the content of the reiterative question that follows was not actually uttered and is not a logical entailment. In the same way as for (128)-(129), the Speaker in (119)-(121) can be construed as checking whether the inference she has drawn is correct, deferring to the superior knowledge of the Addressee.

There is an additional element that becomes more prominent in the latter class of examples, though. In explaining these examples in Section 4.2, I appealed to a process of accommodation to explain how the Contextual Bias Condition can be satisfied. Forcing that accommodation may itself be part of the point of the reiterative question. To the extent the utterance can be accommodated as a question, the Speaker succeeds in establishing that the content expressed is to be regarded as a consequence of the Addressee's public commitments (at least until, and unless, the Addressee refutes it). This is the case even if the Addressee had no intention of conveying the proposition in question with the original utterance, as seems likely in (121), at least. In effect, the Speaker in (121) uses a reiterative question to accuse the Addressee of meaning something by what was said that the Addressee not only didn't literally say but didn't (necessarily) even mean. The same sort of strategy can be employed in less guilt-inducing circumstances as an attempt to clarify the Addressee's original intentions or to draw out relevant implications, as in (119)-(120).

Let us now turn to cases where the content of the question does follow directly from a preceding utterance of the Addressee's. It is in these cases that the full range of Speaker attitudes, from acceptance to implied disbelief, comes into play for rising declaratives and interrogatives. (Falling declaratives, as we know, are more restricted with respect to Speaker attitude.) Two relevant examples are repeated below:

(44) [A&B are looking at a co-worker's much-dented car.

A: His driving has gotten a lot better.

B's response:

a. Has it? I don't see much evidence of that.

b. It has? I don't see much evidence of that.

c. It has. #I don't see much evidence of that.

(46) A: Jake's here.

B's response:

a. Is Jake here? Then let's get started.

b. Jake's here? Then let's get started.

c. (Oh), Jake's here. Then let's get started.

In spite of the diversity of Speaker attitudes available for reiterative questions, there is a common element: the sense that the content of the question is news as far as the Speaker is concerned. It may not be particularly surprising news; the Speaker in (46), for example, gives the impression of having expected Jake's arrival. It may qualify as 'news' to the Speaker because it is incompatible with the Speaker's beliefs, as (44) suggests. But it is news in either case. Below I will speculate about how this effect arises for reiterative questions. First, though, I will make some preliminary distinctions between 'news' as I use the term here and certain propositional attitudes.

The observation that news is not necessarily surprising is an important one. I want to deny that reiterative questions generally, and rising intonation specifically, are inherently associated with 'surprise' or 'incredulity', as is sometimes casually assumed. In addition to (46), we have already seen plenty of examples throughout this thesis where attributing surprise to the Speaker is not warranted. Of course, it is not necessarily ruled out either. The point is just that a reading of surprise is not

inevitable. One independent factor that is consistently associated with readings of surprise and/or incredulity is expanded pitch range. The association of expanded pitch range with readings of surprise is not an effect particular to rising intonation, but has been demonstrated in studies of falling intonation on declarative sentences (Ladd and Morton 1997) as well as the ‘rise-fall-rise’ (Hirschberg and Ward 1992). (In a much smaller-scale study, Gunlogson 1998 obtains similar results for the category of rising intonation under consideration here.)

A second point is that being surprised does not entail being skeptical. One may be very surprised by a piece of news without being seriously inclined to doubt it. For example, if my car has been stolen, I will be very surprised when I return to my parking place and find it gone; I may even, at first, find it difficult to believe my eyes, or to trust my memory about where I parked. But after the initial shock I won’t doubt that it is gone. Similarly, a Speaker who conveys that a piece of news just received is very surprising, difficult to believe, even incredible,¹³ does not thereby rule out the possibility of accepting it. The point is that even when reiterative questions do convey surprise, Speakers are not limited to a particular propositional attitude as a result. Surprise is perfectly consistent both with subsequent acceptance of the surprising news and with rejection of it.

The task at hand, then, is not to understand why a particular attitude arises in connection with a reiterative question but to understand more generally where the sense of novelty to the Speaker comes from. Without attempting to give a

comprehensive answer, I suggest that the solution lies in the reiterative nature of the question itself. The Speaker's indication of a need for the Addressee to repeat material already contributed can have this effect no matter how it is formulated:

(146) A: There's a leopard in the living room.

B: What?! / I beg your pardon? / Huh? / Come again?

An impression that the Speaker is startled by or uncertain about the information just provided is heightened when the Speaker actually falls silent and waits for a confirming or disconfirming answer from the Addressee. In many cases the Speaker uttering a reiterative question does not do this, but continues on. In such cases the reiterative question may function as an acknowledgment that the Addressee's utterance has been attended to (even if it is not ultimately accepted).

Nilsenová 2000 discusses the use of acknowledgments in connection with uncertainties about the common ground in discourse, using probabilistic methods for modeling mutual beliefs. Building on work by van Rooy 2000, she connects the utility of acknowledgments with a measure of information or 'surprisal' value of the message acknowledged. This link between acknowledgment and surprise, I speculate, may operate both ways. If a Speaker who has just received an unanticipated piece of news is more likely to acknowledge it explicitly, then a Speaker who wants to convey that a piece of news is unanticipated may exploit the device of acknowledgment to do so.

¹³ It may be relevant to note that as predicates, both *surprising* and *incredible* are factive.

There is one type of example for which the effect of novelty with respect to the Speaker can be derived in a relatively straightforward way. Reiterative questions may be used to question presuppositions, as in (140) (repeated from p. 116):

- (140) A: Maria's husband was at the party.
B's response:
a. Is Maria married?
b. Maria's married?
c. Oh, so Maria's married.

All three locution types have the effect of conveying that the Addressee was mistaken in assuming that Maria's marital status was common knowledge. The Speaker is apparently hearing about the existence of her husband for the first time.

To understand this phenomenon, I want to return to a point raised in the discussion of vacuousness in Section 3.5. As I have assumed throughout this chapter, it is reasonable to think that speakers avoid making vacuous moves, on general principles. It is also reasonable to think, by the same general principles, that an Addressee will avoid interpreting a Speaker's move as vacuous if there is any other possibility. This was suggested in Section 3.5 in connection with example (103), repeated below:

- (103) B: I've just searched the refrigerator and there's absolutely nothing cold to drink.
A: Yeah, I know. We're out of just about everything.
B:
a. #Are we out of beer?
b. #We're out of beer?
c. #(So) we're out of beer.

The observation was that there is a tendency to try to escape the anomalous effects in (103) by interpreting the questions in (103) as nonvacuous, assuming that they refer to a supply of beer stored somewhere other than the refrigerator.

The idea that interpreters tend to seek nonvacuous interpretations connects with the novelty effect of (140) as follows. Recall that a presupposition is satisfied in a context only if it is a joint commitment of the participants. (See (83)-(84), p. 54.) And a locution is defined as vacuous with respect to a context just in case its content is already a joint commitment of the participants (in (101), p. 66). It follows that the only way to interpret the reiterative questions of (140) as nonvacuous is by assuming that the presupposition was in fact not satisfied in the context. If the presupposition is not satisfied, it can only be B's assumptions that are at fault. A has already explicitly committed to the presupposition by virtue of the preceding utterance. The consequence is that to interpret any of the reiterative questions in (140) as nonvacuous, the Addressee must conclude that B was not previously aware that Maria is married.

This reasoning may also help illuminate why falling declaratives are so in need of the additional support provided by inferential markers (*oh, so, I see*, and the like) to be felicitous in these presuppositional cases (as discussed in Section 4.3). The rising declarative and interrogative versions do not express Speaker commitment, and the failure to express commitment is consistent with the implication that the Speaker had and/or has no commitment to the presuppositional content. With the falling declarative, which does commit the Speaker, it must be

independently made clear that the commitment has just come about as the *result* of A's utterance. Otherwise, if B's awareness is taken to predate the falling declarative utterance, it becomes very difficult to interpret (140c) as nonvacuous, and the anomaly associated with vacuous moves emerges.

Chapter 5

Conclusion

5.1 *Review of the analysis*

The major goal of this thesis was to arrive at a compositional account of rising and falling declaratives, and to do so in a way that illuminates their use as questions and restrictions on such use. In this section I will recapitulate how that goal is achieved by the analysis and discuss implications of the overall architecture of the proposal.

There are two primary components to the analysis. The first has to do with the non-neutrality, or bias, of declaratives, both rising and falling. I first document the empirical manifestations of that bias (Section 2.2) and show that it co-exists, for rising declaratives, with a lack of commitment to the content of the declarative by the Speaker (Section 2.3). As Section 2.4 points out, there is some tension between the notion of bias, or commitment, and the idea that rising declaratives fail to commit the Speaker. That tension is resolved in this analysis by locating the commitment expressed by a *rising* declarative with the Addressee, while a *falling* declarative commits the Speaker. The notions of bias and neutrality are formalized in contextual terms, using a representation of the discourse context that distinguishes between the commitments of the individual discourse participants. As

explicated in Section 3.1, contextual bias toward some proposition p exists when it is possible for p to be accepted as a mutual commitment of the participants without any non-monotonic revision of existing commitment sets, while at the same time there exists no such smooth course to mutual agreement on $\neg p$. In a context neutral with respect to p , on the other hand, both p and $\neg p$ are equally eligible for mutual commitment – no participant is understood as having a public stand on the issue.

The hypothesis about the differing effects of rising and falling declaratives is implemented via context update semantics in Section 3.2. Declaratives are defined as updating the commitment set of an individual participant by eliminating worlds of which the descriptive content of the declarative is not true. The intonational category specifies whether it is the Speaker's (falling intonation) or the Addressee's (rising) commitment set that is updated. The effect of these definitions is to guarantee that the use of a declarative with descriptive content p , rising or falling, will never result in a context neutral with respect to p . A rising interrogative with content p , by contrast with a declarative, can preserve neutrality with respect to p , if it is uttered in a neutral context. An important observation is that interrogatives can also be used in contexts already biased toward their content; they differ from declaratives in having no potential for *introducing* bias.

The crucial theorems are summarized below:

Theorems

(147) No declarative, rising or falling, is neutral with respect to any context.

(148) No interrogative, rising or falling, is biasing with respect to any context.

The ‘commitment’ intuitively associated with declaratives is defined as a property of declarative locutions:

Commitment as a property of locutions (across contexts)

(149) A locution *L* *expresses commitment* to *p* iff there is no context *C* such that *L* is consistent with *C* and *C* + *L* is neutral with respect to *p*.

The analysis thus fulfills the promise of giving formal substance to the concepts of *bias*, *neutrality*, and declarative *commitment*, and in doing so accounts for the patterns of distribution illustrated in Chapter 2.

The second major component of the analysis is the treatment of questioning in Chapter 4. Before proceeding to the summary of that treatment, however, I want to point out a noteworthy feature of the analysis so far. The account of sentence and locution meaning in Chapter 3 makes no reference to pragmatic categories of use such as questions and statements. According to the definitions, declaratives express commitment by virtue of what they mean, without regard to what they are intended to accomplish in a particular context. If the analysis were to stop at the end of Chapter 3, omitting the treatment of questioning in Chapter 4, it would still offer an explanation of why a declarative, rising or falling, is not acceptable as a neutral question. It’s because a declarative isn’t good as a neutral *anything*. Declaratives are simply not neutral. As I will discuss shortly, this hard-and-fast characterization of declaratives as non-neutral has implications for their use as statements as well.

The problem that remains unaccounted for by the end of Chapter 3 is the issue of how declaratives function as questions, when they do, and what the

intuitive difference between rising and falling declaratives as questions amounts to. The answer given in Chapter 4 is that declaratives function as questions only when their effect on a particular context mirrors in certain important respects the effect of a rising interrogative in the same context. Crucially, to be interpretable as a question in a context, a locution must be *uninformative* with respect to the Addressee's commitment set in that context. The Uninformativeness Condition repeated below states this requirement:

Uninformativeness Condition

- (104) An utterance of a locution *L* is a polar question in *C* only if *L* is uninformative with respect to $CS_{Addr}(C)$.

Interrogatives are uninformative by nature, as the definitions in Section 3.5 spell out, and thus satisfy the Uninformativeness Condition trivially. Declaratives satisfy (104) only when the Addressee is already publicly committed to the descriptive content of the declarative – a contextual requirement that imposes significant restrictions on the distribution of declaratives as questions relative to interrogatives. Section 4.2 demonstrates empirically that this picture is accurate, showing that declarative questions are indeed restricted to contexts of the sort required by (104). A welcome consequence is that the well-known 'echo question' use of rising declaratives turns out to be just a special case reflecting the more general condition on declarative questions.

It is noted throughout the thesis that wherever a rising declarative can be interpreted as a question, the corresponding rising interrogative is also felicitous and equivalent in effect. This correspondence between rising declaratives and

interrogatives plays an important role in the treatment of questioning in two ways. First, it makes available a straightforward definition of what it means to be a *polar question*: it means having certain crucial effects of a polar interrogative, whether those effects are achieved by using an interrogative or by other means, such as using a declarative in an appropriate context. The distributional approach to questionhood put forward in Section 4.3 is rigorous enough to make distinctions between what counts as a question and what doesn't, while retaining enough flexibility to account for the variety of particular intentions, attitudes, and functions associated with 'questioning' in a general sense.

The second important aspect of the correspondence between rising declaratives and interrogatives is that it illuminates the way in which rising declaratives are more 'natural' as questions than falling ones. A rising declarative that meets the Uninformativeness Condition *automatically* has the same effect on the context as a rising interrogative with the same content; this follows from the meanings of rising declaratives and interrogatives as defined in Chapter 3, together with the Uninformativeness Condition. There is thus a natural overlap in effect between rising declaratives and interrogatives in the set of contexts in which the Uninformativeness Condition is met. This overlap is formal in nature. It exists independently of how (or whether) we associate a pragmatic category with utterances within the set, as was done with category of *polar question* in Section 4.3.

For falling declaratives, there is no such formal overlap in effect. Unlike the case of rising declaratives, when the Uninformativeness Condition is met for a falling declarative in a particular context, that falling declarative still has the potential to do something an interrogative does not do, namely commit the Speaker. This difference does not mean that falling declaratives can never function in a questioning way. But it does mean that they are not identical in effect to interrogatives when they do so. The most significant point for the analysis is that the difference leads to additional restrictions associated with falling declaratives as questions. For a falling declarative to be interpreted as a question, the Speaker's commitment must be mutually understood as having its basis in the Addressee's knowledge and authority, not the Speaker's own private evidence. No such restriction applies to rising declarative questions, since they do not similarly commit the Speaker.

In the end, the relative naturalness of rising interrogatives, rising declaratives, and falling declaratives is expressed concisely in distributional (rather than functional) terms. Falling declaratives are interpretable as polar questions in a proper subset of the contexts that allow rising declaratives as questions. And rising declaratives in turn are interpretable as questions in a proper subset of the contexts that allow rising interrogatives. Defining *markedness* in terms of this subset relation in Section 4.4, I arrive at the characterization repeated in (138)- (139):

- (138) A rising declarative is more marked as a polar question than a rising interrogative with the same content.

- (139) A falling declarative is more marked as a polar question than a rising declarative (or rising interrogative) with the same content.

I thus claim to have achieved the goals of illuminating the use of declaratives as questions, of accounting for restrictions on their use, and of doing so with an explicitly compositional analysis that takes both intonation and sentence type into account.

5.2 Intonational meaning, sentence type, and context

In this section I will discuss some implications of the present work for the study of intonational meaning generally and its interaction with sentence type distinctions and contextual factors. In doing so I will comment on several previous analyses of intonational meaning that treat or bear on declarative questions – in particular, Bartels 1997, Pierrehumbert and Hirschberg 1990, and Gussenhoven 1983. I suggest that these accounts have in common the characteristic of attributing to an intonational contrast the kind of differences in meaning that I associate with declarative vs. interrogative sentence type, and I assess consequences of this difference in architecture in light of the empirical generalizations introduced earlier.

As stated at the outset, this analysis differs from most of its predecessors in the intonational literature in several significant ways. The first point is that the data introduced in this thesis, particularly the data dealing with contextual restrictions on declarative questions, are organized to bring out empirical generalizations that

have not been addressed in a systematic way by previous proposals. These restrictions play an integral role in shaping the present analysis. What the data establish beyond a doubt is that rising declaratives are equivalent to neither rising interrogatives nor falling declaratives, but form a distinct category. Distinguishing these three categories – rising declaratives, falling declaratives, and rising interrogatives – requires going beyond a simple binary distinction between rising and falling, ‘question’ and ‘statement’, intonation, however those categories are characterized. We need to appeal to the sentence type dimension as well. Thus the implications of the data lead directly to the second defining feature of this account: the explicitly compositional approach to rising and falling declaratives, whereby both sentence type and intonation make distinct contributions to meaning.

Including both the sentence type dimension and the intonational contrast in the analysis allows for a four-way contrast, as shown in Figure 4:

Figure 4: Four-way sentence type/intonation classification

	<i>Rising</i>	<i>Falling</i>
<i>Declarative</i>	Rising declarative	Falling declarative
<i>Interrogative</i>	Rising interrogative	Falling interrogative

I have not discussed the fourth cell, falling interrogatives, and will not do so now; see Section 5.3.2 for some further remarks.

For now I want to point out that a strictly intonational approach to declarative questions – by which I mean any approach in which the sentence type is barred from playing a role, and all the work is done by intonational elements – is handicapped at the outset by not having enough formal distinctions to delineate the necessary number of categories. I doubt that anyone would make an argument for deliberately excluding sentence type considerations from the analysis of declarative questions. Rather, the task of sorting out their formal contribution has simply been neglected (with the exception of Bartels 1997). Independently of the specifics of the proposal I advance, the data presented in this thesis constitute an argument for ending that neglect, putting to rest any simplistic characterization of ‘question intonation’ on declaratives. The encouraging news is that attending to sentence type distinctions and identifying relevant contextual restrictions, far from increasing the complexity of the problem, has the potential to make the notoriously difficult task of characterizing intonational meaning more tractable by isolating the effects that truly belong to intonation.

Another characteristic that sets apart the present account of declarative questions is the crucial role played by the context. The particular features of the context that matter are identified and enlisted in telling the story of how declaratives function as questions. The result is to lessen the burden of explanation for the intonational component, particularly the rise, which does not have to do all the work of ‘turning a declarative into a question’. Declaratives do not ‘turn into questions’ at all on this account. When functioning as questions, they achieve this

effect by means of the ordinary meaning associated with their sentence type and intonational contour, together with constraints imposed by the context on their interpretation.

This three-dimensional approach – involving intonation, sentence type, and context – is what allows the intonational contrast between the fall and the rise to be cast in relatively abstract terms. In this thesis, the role played by rising vs. falling intonation is to attribute the sentence update – declarative or interrogative – to the Addressee vs. the Speaker. The broad Addressee/Speaker distinction attributed to the intonational contrast does not by itself do much to explain the particular distributional patterns associated with declarative questions, or for that matter, how declaratives function as questions in the first place. This is where the explicit reliance on context in the present account is crucial. As Chapter 4 made clear, the absence of a direct link between rising intonation and questioning is actually an advantage, given the otherwise mysterious distributional restrictions on declarative questions. The view defended here is that rising intonation is only part of the picture of declarative questioning. The contextual restrictions are expected once we accept that rising intonation is not sufficient for questioning – even rising declaratives need the proper sort of contextual support to operate as questions.

As was mentioned in Section 1.3, there exist precedents in the intonational literature for the kind of intonational contrast I propose here, the general contrast along the Speaker/Addressee dimension – e.g., Steedman 2000, Bartels and Merin 1997. The present proposal represents an attempt to fully work out the

consequences of this sort of approach for the empirical domain of declarative questions, bringing into play the factors of sentence type and context as well as the intonational contribution. To the extent this attempt is successful, it shows that attributing this sort of general and abstract meaning to intonational elements, rather than linking them tightly to specific discourse functions, is a viable strategy, provided the other key contributing factors are taken into account.

Let us turn now to the sentence type distinction. In this thesis, I have implemented the declarative vs. interrogative contrast in terms of commitment vs. non-commitment, formalizing the relevant notions in Chapter 3. While this aspect of the approach to declarative questions is to my knowledge novel, there do exist related proposals in the intonational literature – related in the sense of attributing to intonational elements the kind of commitment/non-commitment dichotomy I have assigned to the declarative/interrogative contrast. For example, Bartels 1997 identifies the phrasal tone L- (using a Pierrehumbert-style system with some variations) as coding ‘assertiveness’, while the lack of assertiveness associated with the absence of L- gives rise to the implicature of Speaker uncertainty. A speaker with the attitude of ‘assertiveness’ with respect to a proposition and an addressee “expresses an instruction to the addressee to commit himself publicly...” [p. 177], assuming a Stalnakerian representation of the context. Gussenhoven 1983 hypothesizes that with a fall, the Speaker ‘adds’ to the background of the discourse, while a rise is associated with ‘(relevance) testing’. The opposition between these two categories, he comments, is the opposition between ‘putting in’ (the fall) and

‘not putting in’ (the rise). In an obvious way this contrast resembles the notions of commitment and non-commitment as implemented in this work. Pierrehumbert and Hirschberg 1990, adopting Gussenhoven’s suggestions about the kind of meaning associated with intonational elements while rejecting his phonological framework, describe a similar contrast in comparing L* vs. H* pitch accents: H*, it is said, marks the accented element for addition to the Common Ground, while L* excludes the accented element from the predication.

Abstracting away from the details of these proposals for the moment, and from the substantial differences in assumptions about phonological categories and scope, there is a strong family resemblance between the three sets of suggestions just outlined for the treatment of the relevant intonational contrast, considered as a group, and the proposal I have made for the treatment of the sentence type contrast. In the intonational proposals, the element playing the role of what I call the ‘fall’ (★HL for Gussenhoven, L- for Bartels, H* for Pierrehumbert and Hirschberg) is associated with addition to the background or context, while declarative form accomplishes that task in my proposal. Similarly, the intonational approach associates an element corresponding to my ‘rise’ (★LH for Gussenhoven, H- for Bartels¹⁴, L* for Pierrehumbert and Hirschberg) with the discourse operation of presenting a proposition *without* adding it, an operation that resembles the identity function characterizing lack of commitment in my treatment of syntactic

¹⁴ This is a simplification. Bartels does not assign meaning to H- but the absence of L- (which amounts to the presence of H-) has systematic effects on meaning.

interrogatives. In the discussion that follows, I will use the term ‘commitment/noncommitment’ as a way of referring to the general kind of contrast common to these proposals as well as my own (without invoking the specific implementation associated with these terms in my analysis). I will also attempt where possible to refer to the relevant intonational contrast in a neutral way by using terms such as ‘commitment’ intonation (instead of ‘fall’) and ‘noncommitment’ intonation (replacing ‘rise’).

What are the consequences of locating the commitment/noncommitment dimension of contrast in the system of intonational meaning as opposed to associating it with the sentence type, as I do here? The first challenge that arises for this alternate approach, in view of the data presented in this thesis, is formulating how the commitment vs. noncommitment contrast signaled by the intonational element plays out on declaratives vs. interrogatives. In other words, if commitment/noncommitment goes with the intonational contrast, how is the contrast between declarative and interrogative form to be characterized? Recall from the discussion earlier in this section that the option of ignoring sentence type is not viable, given that rising declaratives must be distinguished from rising interrogatives (and falling declaratives from falling interrogatives).

Furthermore, the contrast between declaratives and interrogatives must be characterized in such a way as to allow either type compatibility with either intonational element. The canonical combinations of ‘commitment’ intonation with declaratives and ‘noncommitment’ intonation with interrogatives would seem to

present no particular problems, since whatever the sentence type is hypothesized to do, it seems likely to mesh with the intonational choice in these cases. But the combination of a declarative with the intonational element associated with ‘noncommitment’ presents a more significant challenge.

This point has already come up in connection with the empirical generalizations presented in Chapter 2, repeated below:

- (8) Declaratives [rising or falling] express a bias that is absent with the use of interrogatives; they cannot be used as neutral questions.
- (9) Rising declaratives, like interrogatives, fail to commit the Speaker to their content.

As I pointed out in the discussion of (8) and (9) in Section 2.4, for rising declarative questions there is a certain tension between the requirements imposed by these two sets of observations. The bias that accompanies rising declarative questions (generalization (8)) must be formulated in such a way as to remain compatible with the lack of Speaker commitment (generalization (9)). Recall that my analysis resolves this tension in the following way. Declaratives are characterized as invariably expressing commitment (therefore resulting in bias), in the manner detailed in Chapter 3. A rising declarative does not commit the *Speaker*, however, because the rise attributes the commitment to the Addressee. Only falling declaratives commit the Speaker.

What I am calling the intonational approach to the commitment/noncommitment distinction has no difficulty with (9): by hypothesis, the relevant intonational element encodes lack of commitment, or something akin

to it, directly. But how is the bias of declaratives described in (8) to be represented, and how is declarative bias to be reconciled with the absence of commitment? This is not a point specifically addressed by any of the proposals mentioned (though Bartels comments in passing (p. 395) that the proposition presented in a rising declarative question is ‘posed’ as a rationally justified inference rather than a mere guess). In the absence of a specific proposal, I will simply identify this as an open issue for the intonational approach. The architecture I have adopted, on the other hand, provides a solution – and indeed, was designed as a response to the empirical observations.

The second set of issues for the intonational approach revolves around the contextual prerequisites for declarative questions, i.e., the Contextual Bias Condition. Recall that under my analysis, declarative questions are restricted to certain contexts because those are the only contexts in which the requirements are met for their interpretation as questions. Rising intonation on a declarative does not directly signal lack of commitment or questionhood; rather, it just attributes commitment to the Addressee. This will only result in a question interpretation if the context is of the right sort. The connection between rising intonation and questioning uses of declaratives does exist, but is not intrinsic to the rise – rather, it is an effect due to a combination of factors that emerges only in the presence of the appropriate sort of contextual support, thereby providing an explanation of the restrictions.

Analyses that make a more direct link between the presence of a particular intonational element and the interpretation of a declarative as a question are at a disadvantage in accounting for the distributional restrictions. Paradoxically, the more successful an account is in tracing questioning uses of declaratives to the contribution of a particular intonational element, the more awkwardness it faces in accounting for contextual prerequisites for such uses. If the presence of the intonational element suffices for the functioning of a sentence as a question, then why should the context have to be in a particular configuration? This problem arises for the intonational approach to commitment/noncommitment to the extent that the presence of an intonational element indicating noncommitment (in its various proposal-specific forms) is understood to directly facilitate questioning interpretations of declaratives. In Bartels's proposal, which is the most explicit of the three, there does seem to be a fairly direct connection between the presence of H-/absence of L- and questioning. But once again, since the relevant contextual restrictions are not specifically addressed in any of these accounts, I will simply note that the restricted distribution of declarative questions relative to interrogative ones, which is a consequence of my proposal, does not follow in any obvious way under the intonational approach.

5.3 *Future developments*

The analysis of declarative meaning I have defended in this thesis was motivated by facts concerning the distribution of declaratives as questions, and in particular, by restrictions on the use of declaratives as questions. However, there are implications as well for other uses of declaratives, as well as for other sentence types and intonational categories. I will close by mentioning some of these implications and identifying open issues.

5.3.1 *Declaratives as statements*

The canonical use of *falling* declaratives is generally considered to be stating or asserting. What can be said about this canonical use and how it relates to locution meaning in the present proposal? In any case, the account of locution meaning given in Chapter 3 stands unchanged. The claim is that declaratives express commitment by virtue of being declarative, regardless of the use to which they are put (and regardless of their intonation). The use of a declarative cannot, by definition, result in a context that is neutral with respect to the descriptive content of the declarative. Falling declaratives have the additional property of invariably committing the Speaker to their content.

Both of these properties of falling declaratives – non-neutrality or ‘bias’, and Speaker commitment – are compatible in intuitively obvious ways with functions such as stating. The task of making this intuitive connection precise would require

developing a careful treatment of ‘statement’ and/or ‘assertion’ in pragmatic terms to parallel that developed for ‘polar question’ in Section 4.3. Classifications based on formal properties of the locution and its interaction with the context, like those introduced in Section 3.5, can be employed in this task. For instance, just as *questioning* uses were hypothesized to require uninformativeness with respect to the Addressee, so uses characterized as *telling* might be hypothesized to involve informativeness with respect to the Addressee.

It should be perhaps be noted that the update meaning associated with falling declaratives in this analysis is not equivalent, and cannot be equivalent, to the common formal treatment of assertion, following Stalnaker 1978, as addition of the asserted proposition to the Common Ground (or equivalently, reduction of the context set). Using the divided context developed here, that notion of assertion amounts to making a proposition as a *joint* commitment of the participants. Under the present analysis, a falling declarative by itself cannot accomplish the effect of assertion, so defined. That is because a falling declarative is defined to operate just on the Speaker’s commitment set, not on joint commitments.

Of course, it may be the case that the fact of the Speaker’s commitment suffices to bring about the Addressee’s commitment as well. This may, indeed, be exactly the effect the Speaker is seeking. Having an assertive *intention* is certainly compatible with use of a falling declarative, given an appropriate context. But unlike most context update accounts of declaratives to date, this analysis does not equate the meaning of the declarative to its assertive use. The advantage of this

approach is that the many uses of falling declaratives that do not clearly fall under the heading of ‘assertion’ as commonly defined (e.g., agreeing, denying, insisting) are included in the story. Falling declaratives are expected to be suitable for any use with which their properties of bias and Speaker commitment are compatible. Thus, just as for questioning, the properties associated with the form of the locution constraint the uses to which it can be put without wholly determining its use. Context will play a role here as well as in questioning. Although the general shape of the approach can be discerned, these suggestions remain at a programmatic level and await implementation.

What about rising declaratives as statements? The issues in this case are more complex. Like falling declaratives, rising declaratives have the property of non-neutrality that goes with declarative form. This property, which interrogatives lack, seems to be a minimal requirement for candidacy as a statement, as was noted in connection with examples like (32), repeated below:

- (32) a. #Is my name Carl? #Will I be your waiter tonight?
b. My name is Carl? I’ll be your waiter tonight?
c. My name is Carl. I’ll be your waiter tonight.

In spite of their informative potential as illustrated in (32), rising declaratives are intuitively not as prototypical in statement use as their falling counterparts. It is true that for many speakers rising declaratives are a perfectly natural way to convey new information. But it is also true that there are many speakers for whom this use is not readily available. Falling declaratives as statements, by contrast, are possible for all speakers. The situation for rising declaratives as statements can also be

compared to the use of rising declaratives as questions. As far as I know, the questioning use of rising declaratives does not exhibit the same kind of variability across speakers that the informative use does, but is available across dialects. Questioning uses of rising declaratives seem to be more fundamental than informative uses in this respect. Furthermore, questioning uses exist side-by-side with informative uses for speakers who have both – the informative use does not replace the questioning use.

What we would like to understand, then, is both how rising declaratives can function in a statement-like way and why they do not have the same status as falling declaratives even when they are able to do so. The present proposal offers no solution to these puzzles, but it does suggest a line of investigation – namely, locating both the potential for functioning as a statement and the contrast with falling declaratives in availability of such use with the locutionary effect of rising declaratives. That effect is to commit the Addressee rather than the Speaker to the content expressed. The speculation in very general terms is that such a move *can* be connected with the function of stating or informing, but that the connection is not as direct as for falling declaratives.

As has already been pointed out, the abstract meaning associated with rising intonation in this proposal does not lead to an inevitable association with questioning. Thus, the door is open for an account of non-questioning uses of rising declaratives. The burden of such an account is to explain how attribution of commitment to the Addressee can work as a way of informing the Addressee, and

how exactly that move differs from the strategy of Speaker commitment expressed by a falling declarative. This work remains to be done.

One final point can be made. Recall that the ‘naturalness’ of rising declaratives as questions was traced in Section 4.3 to their overlap in distribution and effect with rising interrogatives. This overlap, it was pointed out, stemmed from the formal properties of the locutions involved and their interaction with the context, and thus stands as a consequence of the proposal. When we look at falling declaratives vs. rising declaratives, no such overlap obtains. That is, a rising declarative cannot achieve the effect of a falling declarative just by being uttered in an appropriate context (leaving aside vacuous utterances). To achieve the effect of falling declarative, a rising declarative would have to commit the Speaker – and this it expressly does not do. The significance of this point is that we do not expect a *natural* overlap in function between rising and falling declaratives, either – natural in the sense of following directly from formal properties. If rising declaratives do manage to carry out some of the functions that falling declaratives can perform, it will be by a more circuitous route – involving, I assume, the development of novel conventions of usage, as Bartels 1997 suggests in discussing the same category.

5.3.2 Interrogatives

As was pointed out in Section 5.2, the two dimensions of contrast considered in this thesis – rising vs. falling intonation, declarative vs. interrogative form – lead

to a 4-way classification: rising and falling declaratives, rising and falling interrogatives. Of these four locution types, I have discussed three, systematically ignoring the category of falling interrogatives. This neglect is not principled but is the result of the focus on declaratives. Concentrating on declaratives, I have not offered a full account of interrogatives, but have simply identified one of their (negative) properties – they do not, according to the analysis in Section 3.3, make substantive changes to the commitment sets of either discourse participant. The difference between rising and falling interrogatives in the account as it stands is not very significant: a rising interrogative represents an identity function on the Addressee’s commitment set, whereas a falling one would represent an identity function on the Speaker’s commitment set. The effect in both cases is to leave the context unchanged. No predictions follow from this.

Clearly polar interrogatives deserve an account on the same level as declaratives, with attention to the interactions of interrogative form with intonation and the resulting effects on distribution across contexts. (For that matter, so do *wh*-interrogatives and alternative questions; see Bartels 1997 for discussion.) A prerequisite for extending the present approach from declaratives to interrogatives is a non-trivial update semantics for interrogatives, which in turn requires a richer representation of the context, one that goes beyond just commitment to include structures appropriate for modeling interrogatives. As mentioned, there are already candidates for this role – Groenendijk 1999, Büring 1995 and Roberts 1996. What is not clear without further investigation is how the richer notions of context put

forward in these proposals, and the update semantics that goes with them, can best be merged with the divided context argued for here.

In short, I have said something about the commitment associated with declaratives, and the effects of attributing that commitment to the Speaker vs. the Addressee. The appropriate notion to parallel ‘commitment’ for interrogatives, together with characterization of the effects of assigning it to different participants, await further development.

5.3.3 *Bias and neutrality cross-linguistically*

The present study is confined exclusively to English, and there is no expectation that the meanings proposed for individual elements in English will be exactly the same from language to language. In particular, I do not claim that rising and falling intonation have the same meaning in every language (or even that the corresponding intonational categories exist), or that the contrast between declaratives and interrogatives plays out in exactly the same manner. Indeed, some languages lack the kind of syntactic distinction that defines the declarative/polar interrogative contrast in English. In such languages the division of work between syntactic devices and intonational ones in marking categories relevant to questioning may well be allocated differently. I have not investigated such differences and similarities cross-linguistically, and there are doubtless important issues that will emerge in such investigations.

What I do expect, *a priori*, is that all languages will have devices to distinguish between neutral (or potentially neutral) polar questions and questions that introduce or assume bias. In English, declarative questions are only one instance of this latter category. Negative polar interrogatives and tag questions immediately spring to mind as further instances. Cross-linguistically, particles often convey shades of meaning that can contribute to a biasing effect, and that are often as difficult to capture as the nuances of intonational meaning. Beyond the work to which it is put here in the study of intonational meaning and sentence type, I hope that the model of contextual bias and neutrality outlined in Chapter 3 can be useful in studying these and related phenomena cross-linguistically.

5.4 In closing

In this thesis I set out to give a compositional account of rising and falling declaratives, focusing on their use as questions and referring to a body of observations that illustrate restrictions on such use. The picture that emerges is one in which intonational and sentence type meaning constrain how utterances with particular content function in discourse but do not determine their function. Understanding the questioning use of declaratives, I have argued, does not reduce to a problem of knowing when to assign ‘question force’ but requires a deeper investigation of the complex interaction between context and elements of linguistic

form that speakers can exploit to achieve their purposes. Declaratives as questions remain true to their declarative form.

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